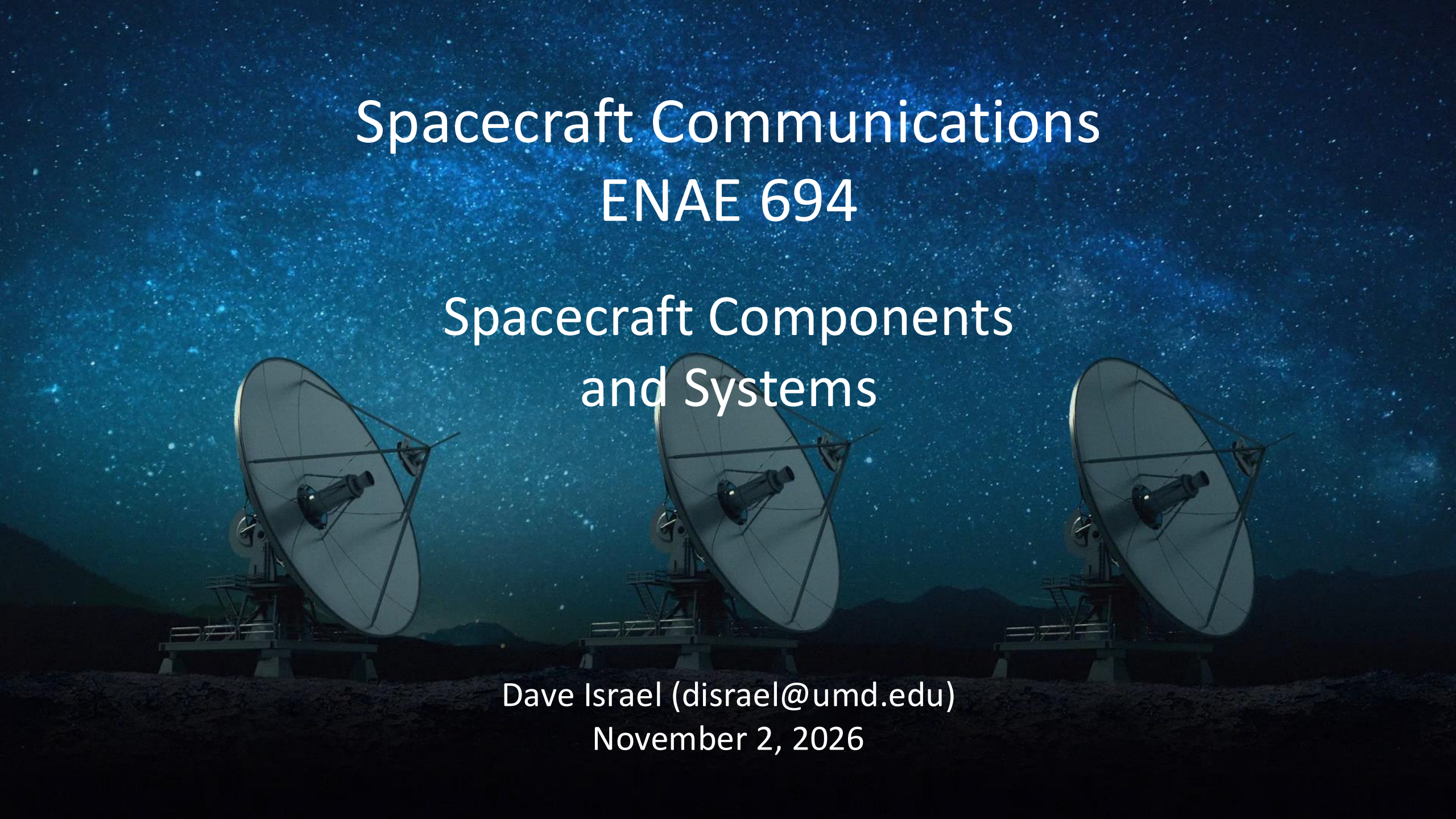


Spacecraft Communications
ENAE 694

Spacecraft Components and Systems

Dave Israel (disrael@umd.edu)

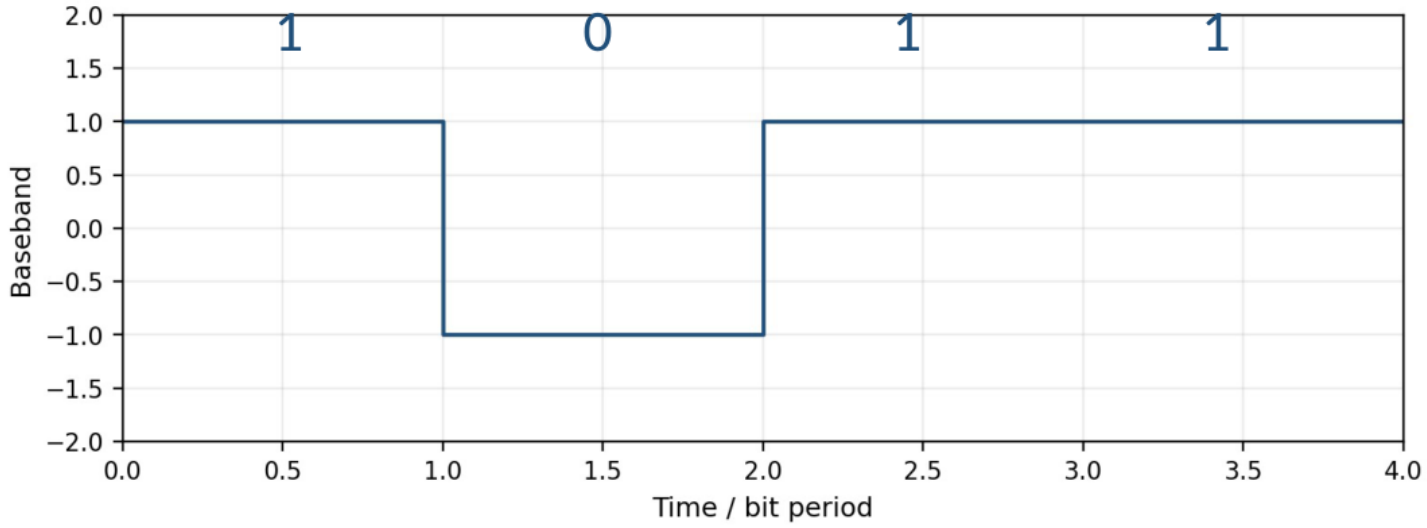
November 2, 2026

Tonight’s Class

Part	Focus	Outcome
1	Components and RF paths	Explain what each component does
Break		
2	Redundancy and complete systems	Trace normal and failed operating modes

The hardware diagram and the link budget must describe the same system.

Review from Last Week: Integrate, Then Decide



For each bit interval

Integrate the signed baseband value.

Compare with zero.

Reset for the next bit.

Average over each aligned bit: +1 −1 +1 +1

Recovered bits: 1 0 1 1

*Keep the sign. Squaring the baseband first would erase the bit distinction.

[12] • Ideal rectangular BPSK after carrier recovery.

Correction from Last Week: Coding and Error Checks

Term or result	Meaning
Information bits / coded bits	Coding adds transmitted bits, not new information
Modulation symbol	One selected waveform state per symbol interval
RS(255,223) symbol	One 8-bit byte
Zero syndrome	The checks pass; an undetected error is still possible

A single parity check detects an odd number of bit flips.

[12]

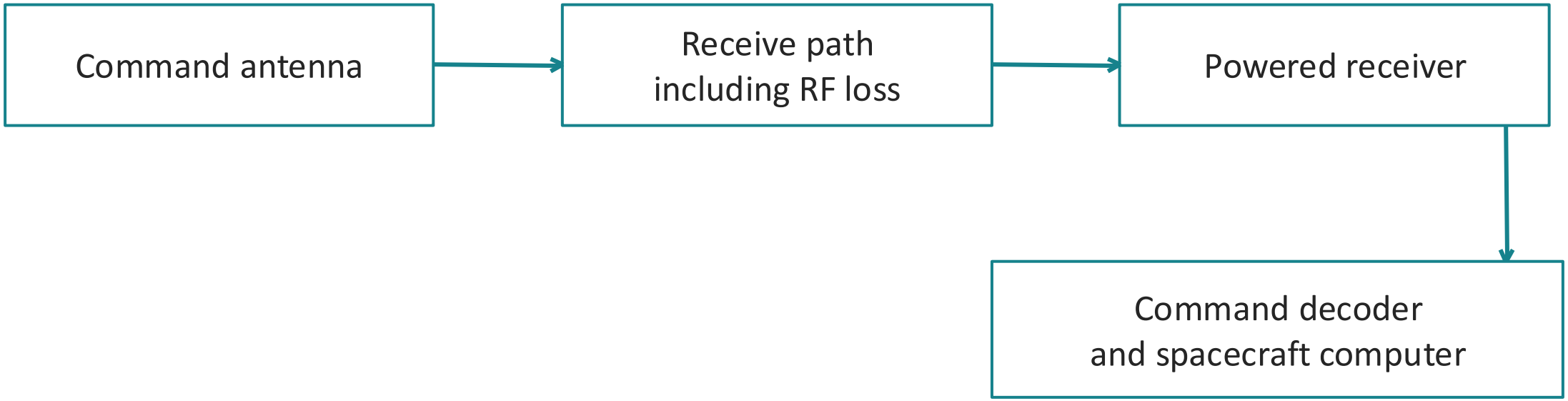
New Material: The Onboard Communications System

Which radios, amplifiers and antennas must fly?

How do RF paths change with operating mode?

What remains usable after a specified failure?

A Command Receiver Must Be Available and Listening



Keep a usable command path through every operating mode.

Ready to listen does not guarantee continuous Earth visibility or network access.

NASA/JPL onboard systems and fault protection [1, 10].

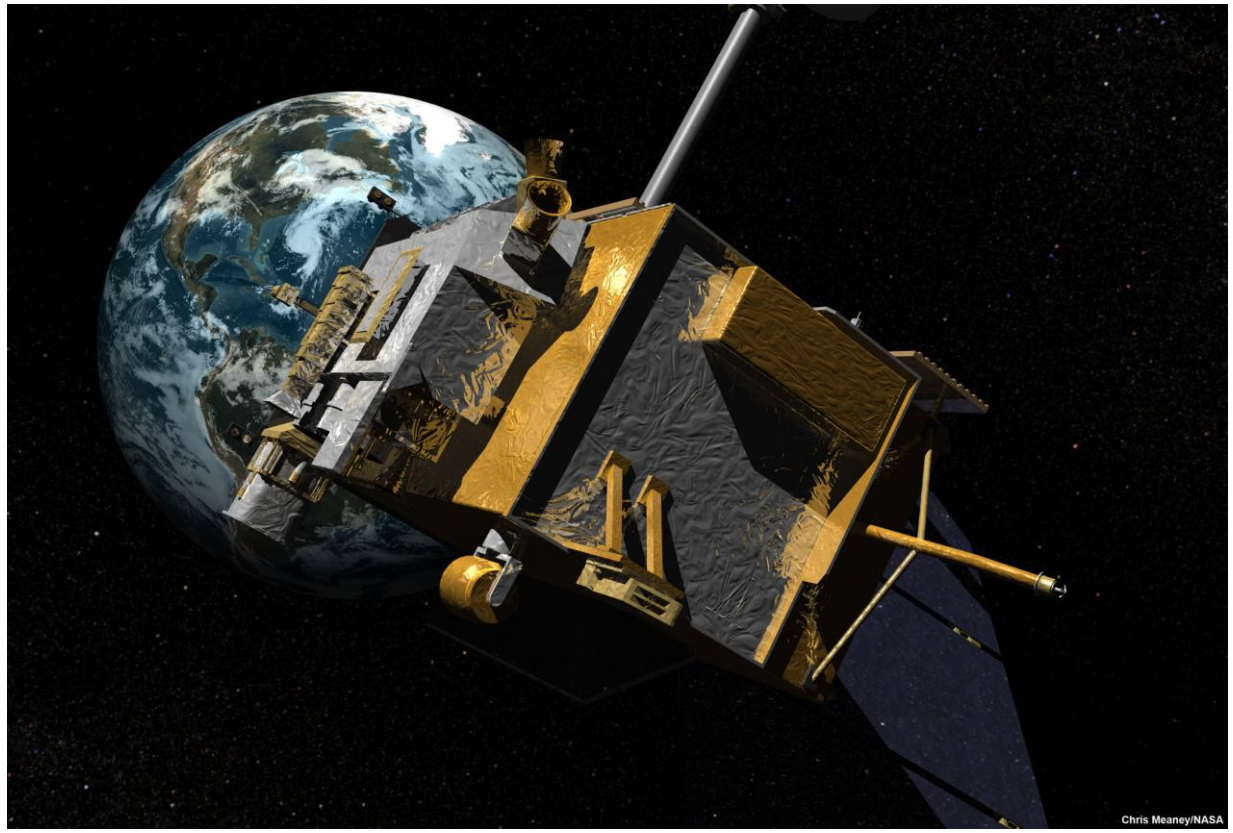
The Design Starts With Operating Modes

Mode	Required communication	Likely antenna need
Science return	High data rate to Earth	Pointed high-gain antenna
Routine command	Reliable command reception	High-gain or broad-coverage path
Safe mode	Low-rate command and telemetry	Coverage with limited pointing
After a radio failure	Recover command and basic telemetry	A usable alternate RF path

A science link alone does not satisfy the spacecraft's communications needs.

Teaching example

Real Hardware Occupies Real Places



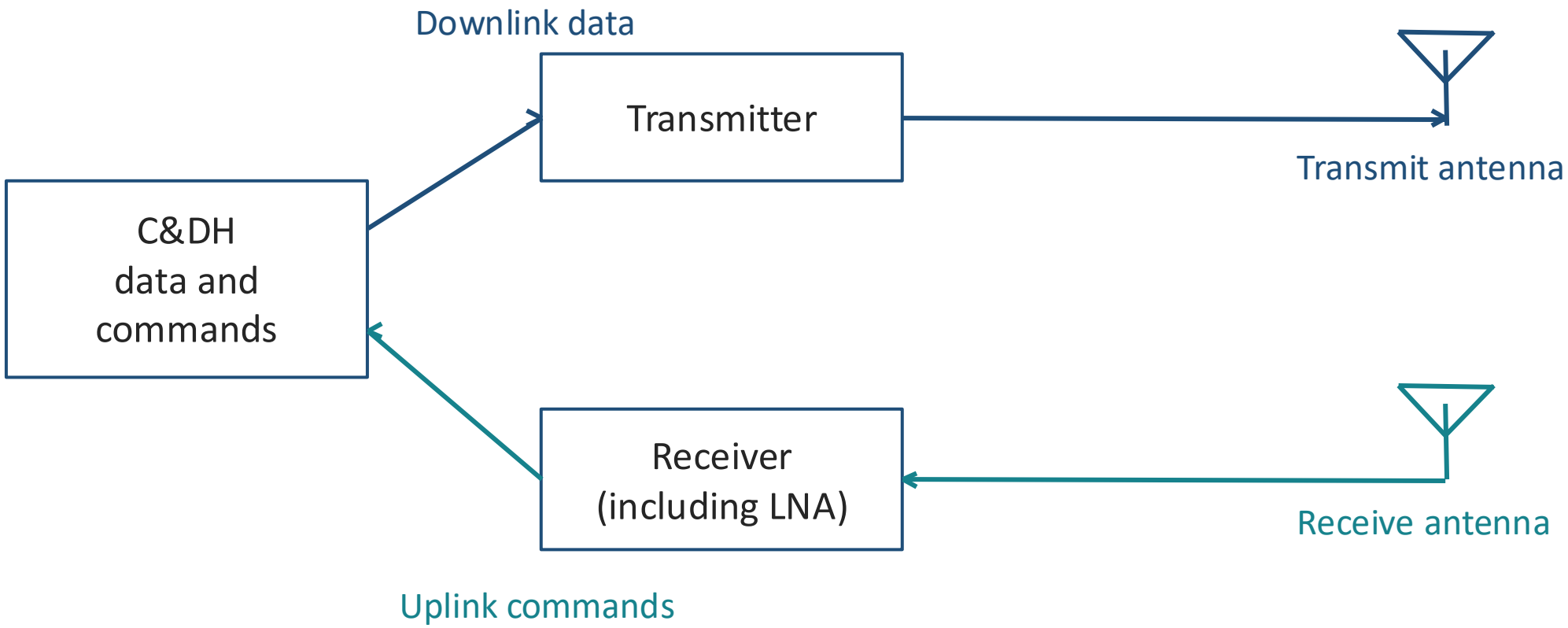
Antennas need a clear view.

Radios and amplifiers need power and thermal paths.

Cables connect the locations and introduce RF loss.

Illustration: Chris Meaney/NASA [4]

Example A: Separate Transmit and Receive Paths



Digital interfaces connect the radio to the spacecraft computer.

[1]

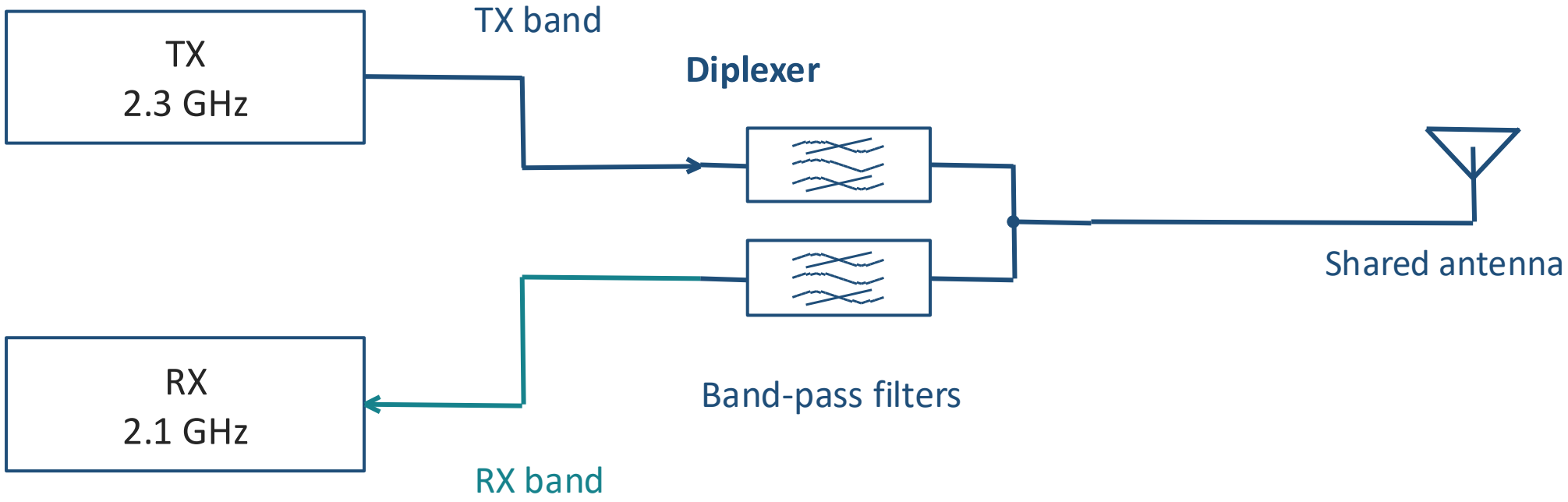
Transmitter, Transceiver, and Transponder

Equipment	What it provides
Transmitter	Data modulation and an RF output
Receiver	RF reception, synchronization and recovered data
Transceiver	Transmit and receive functions in one unit
Transponder	Receives uplink signals and provides a downlink response

The required functions determine what must be inside the box.

[1, 11]

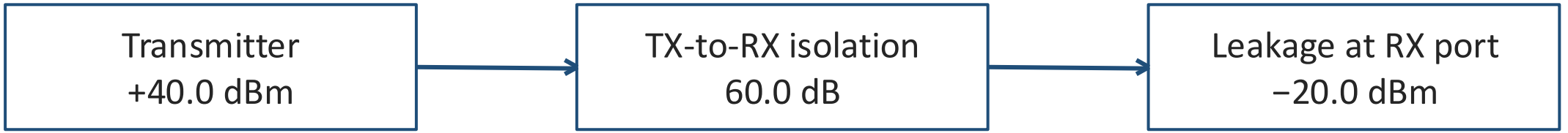
Example A: One Antenna, Two Frequency Bands



A diplexer separates bands. It does not select which radio is active.

[2]

Isolation Protects the Receiver

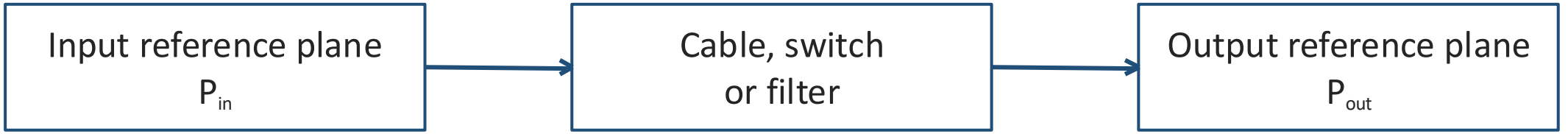


$$[P_{leak}] = [P_{TX}] - \text{isolation} = 40.0 - 60.0 = -20.0 \text{ dBm}$$

Check receiver blocking, compression, damage limits, and TX noise in the receive band. Isolation is frequency dependent.

[2, 3]

Insertion Loss: What Changes When a Part Is Added?



$IL = 10 \log_{10}(P_{before} / P_{after})$

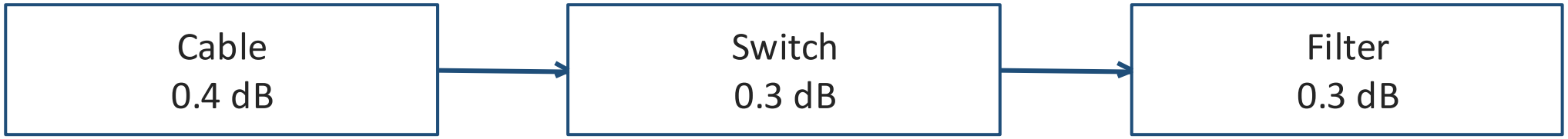
For matched ports: $IL = [P_{in}] - [P_{out}]$

Example: 10.0 W in, 5.0 W out gives 3.0 dB loss.

*Compare the same load and source conditions. Specify frequency and selected ports.

[3, 5] • Matched ports for the input/output form below.

Insertion Loss Along an RF Route



$L_{route} = 0.4 + 0.3 + 0.3 = 1.0$ dB

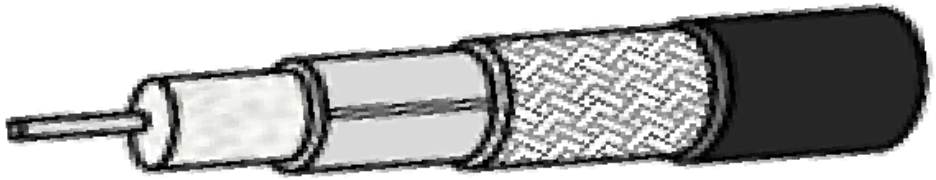
$[P_{out}] = 10.0$ dBW − 1.0 dB = 9.0 dBW

$P_{out} = 10^{9.0/10} = 7.9$ W

A two-way splitter adds 3.0 dB division to each branch, plus any excess loss.

Teaching example • Matched ports at one operating frequency.

Coaxial Cable and Hollow Waveguide





Coaxial cable

Center conductor and shield

Flexible routing, broad frequency use

Loss rises with frequency and length

Hollow waveguide

Fields travel inside the metal guide

Low loss and high power in its band

Bulk, flanges, bends and cutoff matter

Images: Times Microwave [19], Eravant [20].

Cable Loss: Length and Frequency Both Matter

Frequency	Catalog attenuation	Loss in 5.0 m
0.9 GHz	0.13 dB/m	0.7 dB
2.5 GHz	0.23 dB/m	1.2 dB
5.8 GHz	0.36 dB/m	1.8 dB

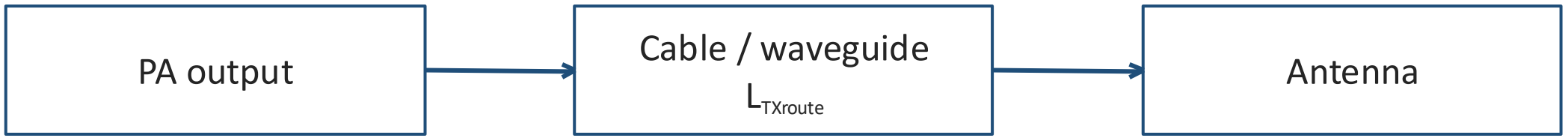
$L_{cable} = \alpha(f) \times length$

At 5.8 GHz: 0.36 dB/m × 5.0 m = 1.8 dB

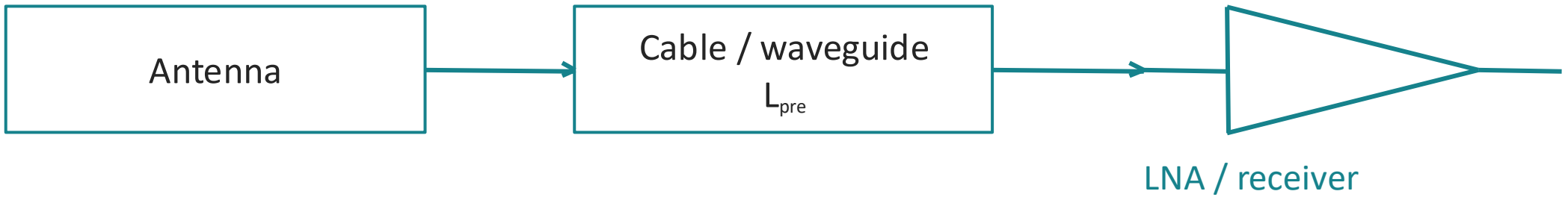
*Add connector and joint losses. Use the selected cable's frequency and temperature data.

[19] • LMR-400 catalog example, not a spacecraft qualification.

Routing Loss: The Location Changes the Consequence



$$[EIRP] = [P_{PA}] - L_{TXroute} + [G_{TX}]$$

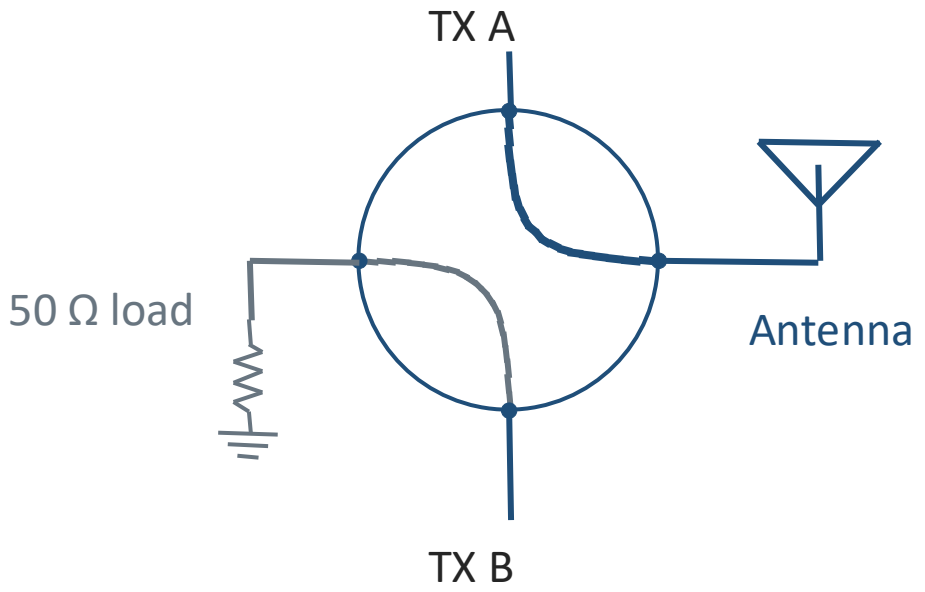


$$NF_{total} = L_{pre} + NF_{receiver} \quad \text{(at 290 K)}$$

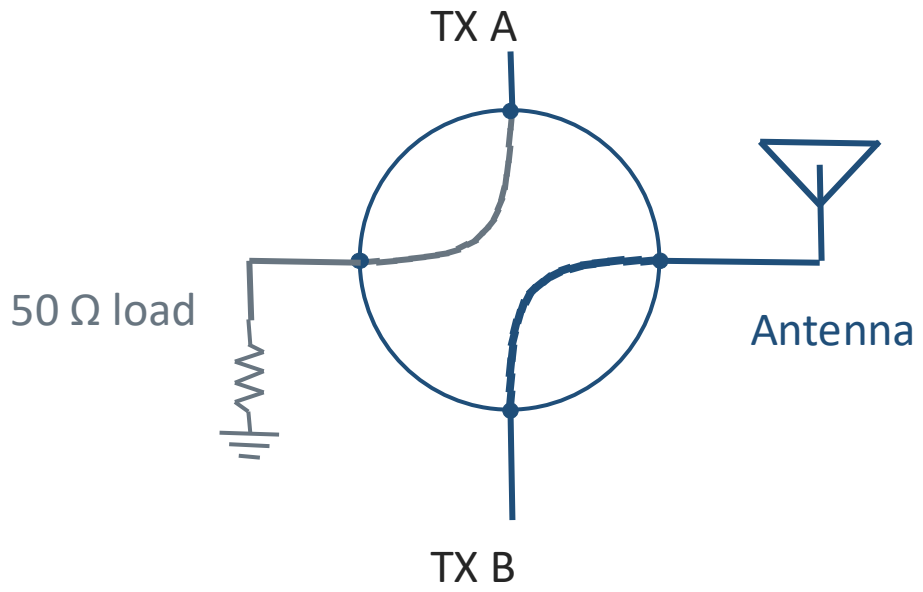
[6] • Passive input loss at 290 K for the NF relation.

RF Transfer Switch: The “Baseball” Symbol

State A: TX A to antenna



State B: TX B to antenna

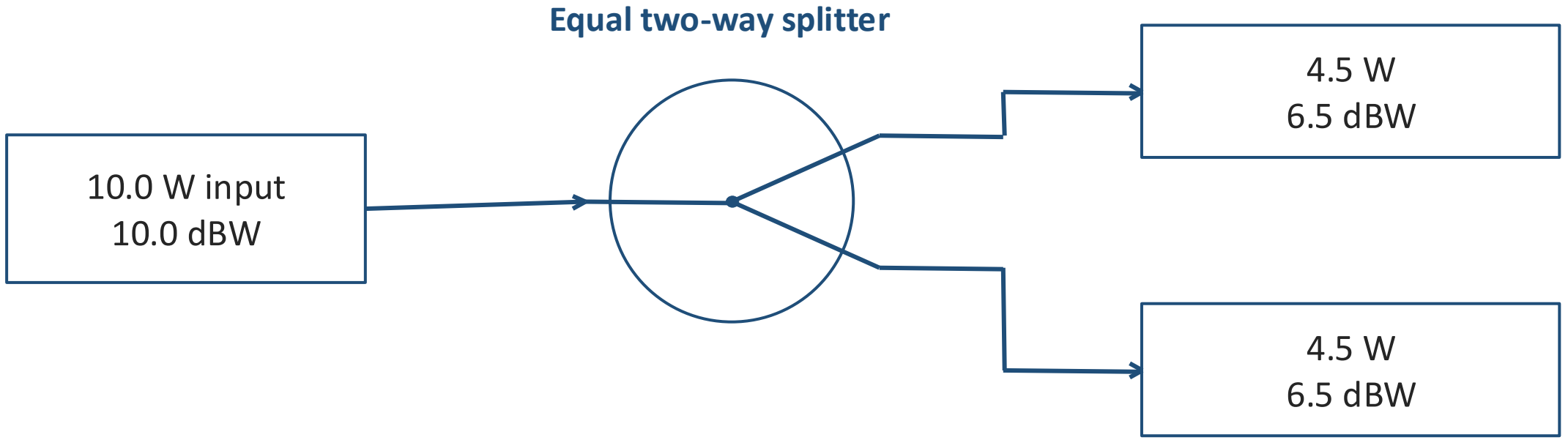


Four ports, two RF paths. Switching changes both connections.

Here the spare TX connects to the load and stays off.

[16, 17] • Connections are reciprocal. Check power rating and switching conditions.

Power Splitter: Division Plus Excess Loss



L_{branch} = 3.0 dB division + 0.5 dB excess = 3.5 dB

$[P_{out}] = 10.0 - 3.5 = 6.5$ dBW per output

[5] • Teaching value: 0.5 dB excess loss. Matched ports.

Example B: Question: Switch or Splitter?

One 10.0 W transmitter must feed the selected antenna.

Design	Path loss	Antenna gain
A: switch	0.5 dB	10.0 dBi
B: equal splitter	3.5 dB total	10.0 dBi

Which gives the higher EIRP toward the selected antenna's target?

*Splitter: 3.0 dB division + 0.5 dB excess loss. Other routing losses omitted equally.

Teaching example

Example B: Worked Answer: Power in the Selected Path

$[P_{TX}] = 10 \log_{10}(10\ W / 1\ W) = 10.0\ dBW$

$[EIRP] = [P_{TX}] - L_{route} + [G_{TX}]$

A: $10.0 - 0.5 + 10.0 = 19.5\ dBW$

B: $10.0 - 3.5 + 10.0 = 16.5\ dBW$

The switch preserves 3.0 dB more EIRP in the selected path.

The splitter buys simultaneous connections at the cost of power per path.

Teaching example

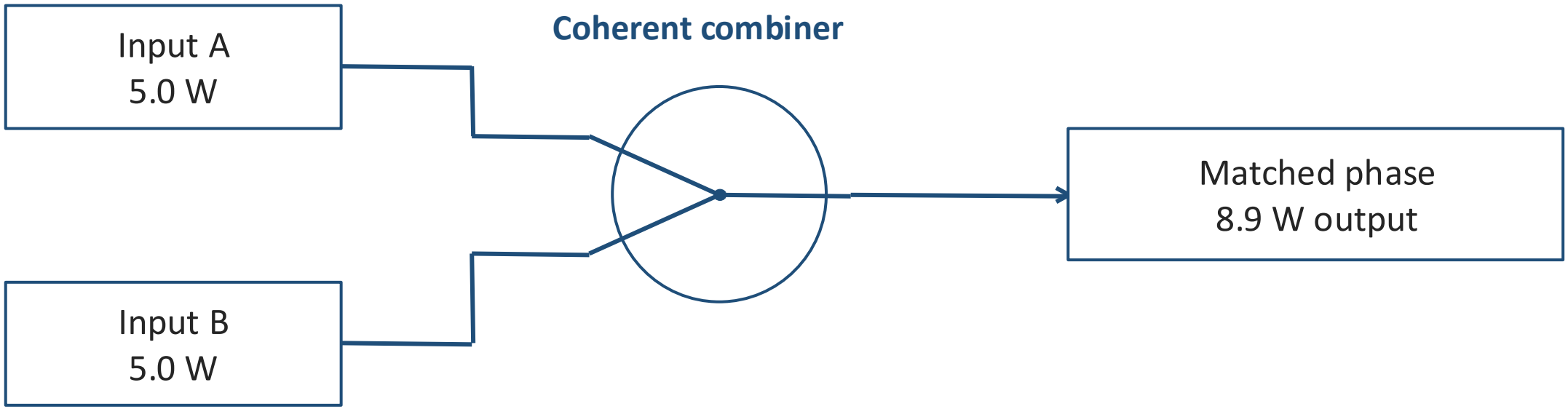
Switch, Splitter, or Diplexer?

Choice	What it provides	Main trade
RF switch	Select a path or paired paths	Low selected-path loss; actuation and failure state matter
Equal splitter / hybrid	Simultaneous connections	3.0 dB division per branch, plus excess loss
Diplexer	Separate different frequency bands	Filter loss and finite rejection; shared common port

Check: required simultaneous paths, isolation, power handling and failures.

[2, 3, 5, 16, 17]

Combining Power: Phase and Frequency Matter



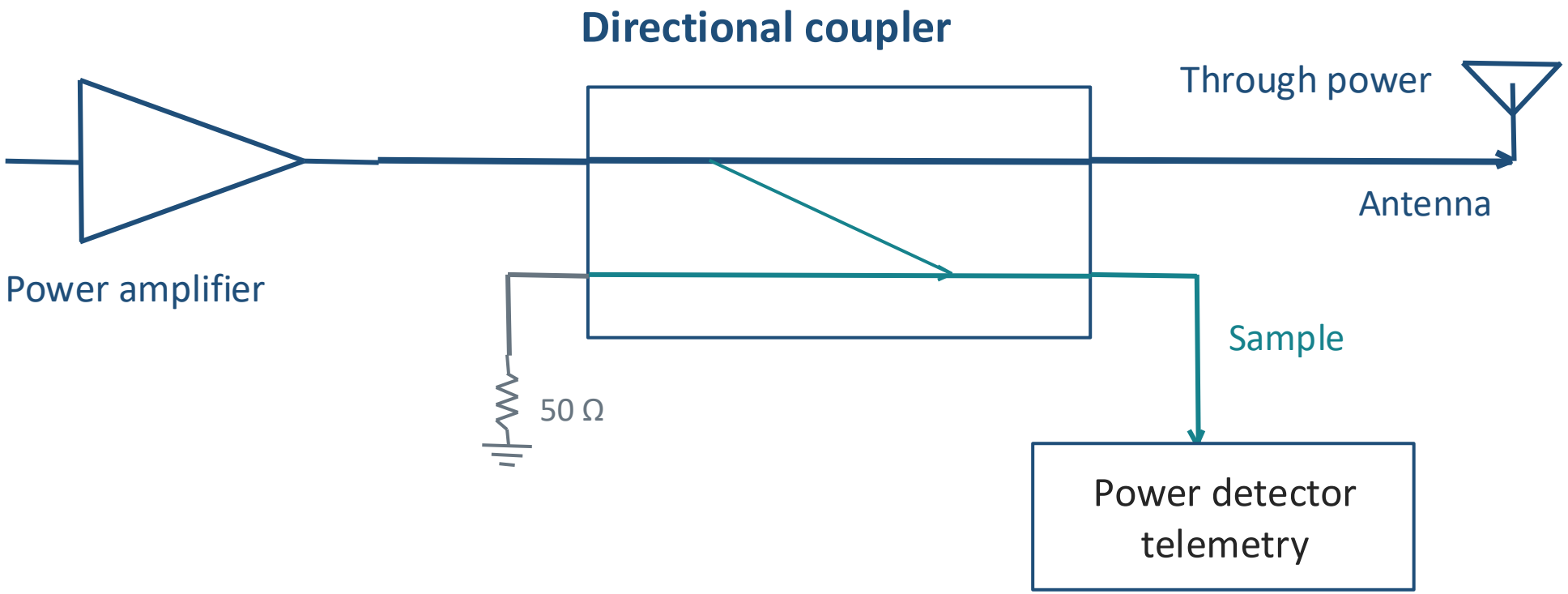
Total input: 10.0 W. Output: 10.0 dBW – 0.5 dB = 9.5 dBW.

Correct relative phase combines both inputs. A diplexer joins different bands.

[2, 5] • Matched coherent inputs. Teaching combiner excess loss: 0.5 dB.

A Coupler Samples Power Without Equal Division

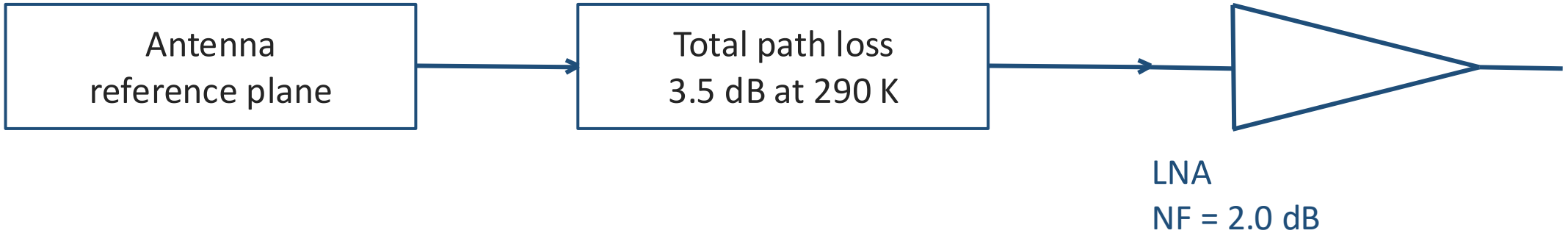
Most RF power continues to the antenna.



The coupled port samples power. It is not an equal split.

[5, 18] • The isolated port has a matched termination.

Receive-Path Loss Changes the Noise Figure



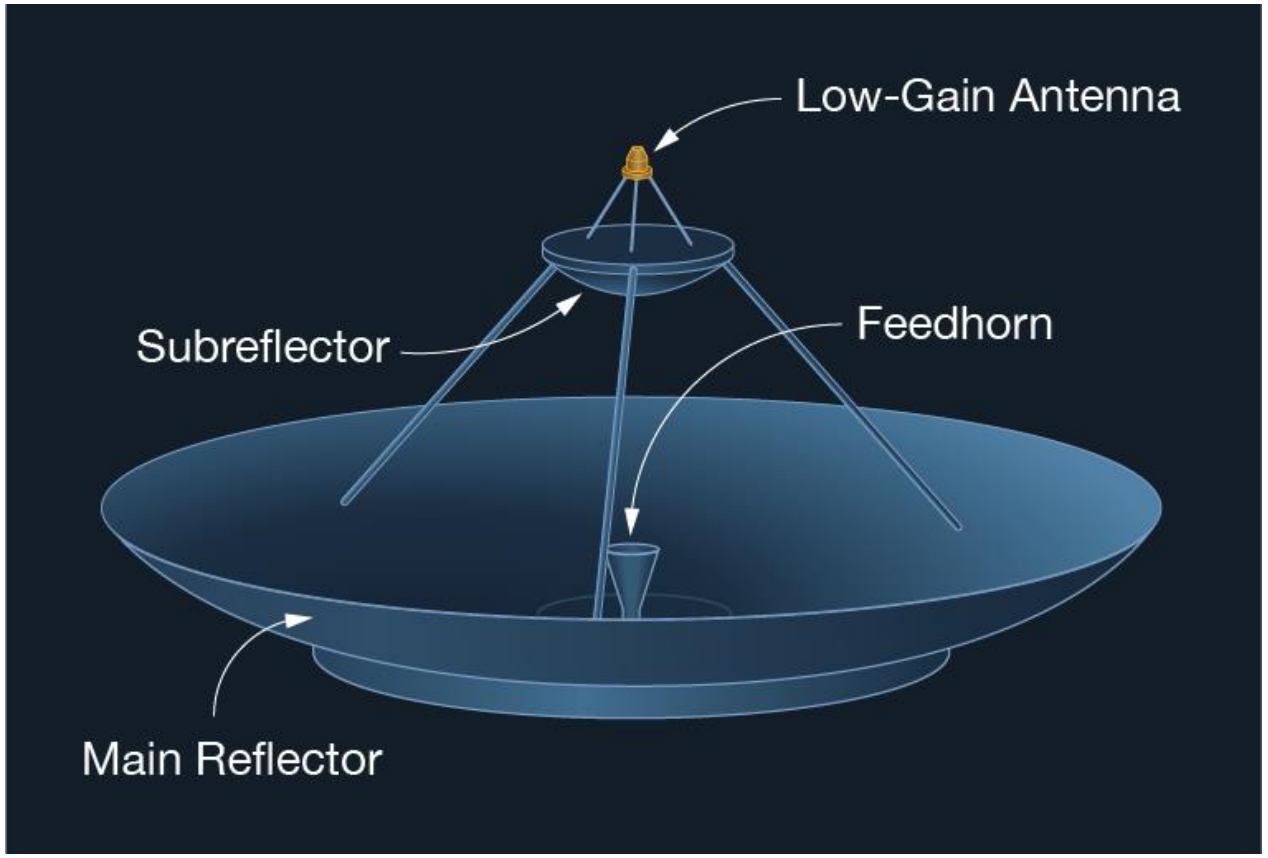
At 290 K: $NF_{total} = L_{passive} + NF_{LNA}$

$NF_{total} = 3.5 + 2.0 = 5.5$ dB

A splitter or cable ahead of the LNA affects receiver sensitivity.

[6]

Multiple Antennas Serve Different Conditions



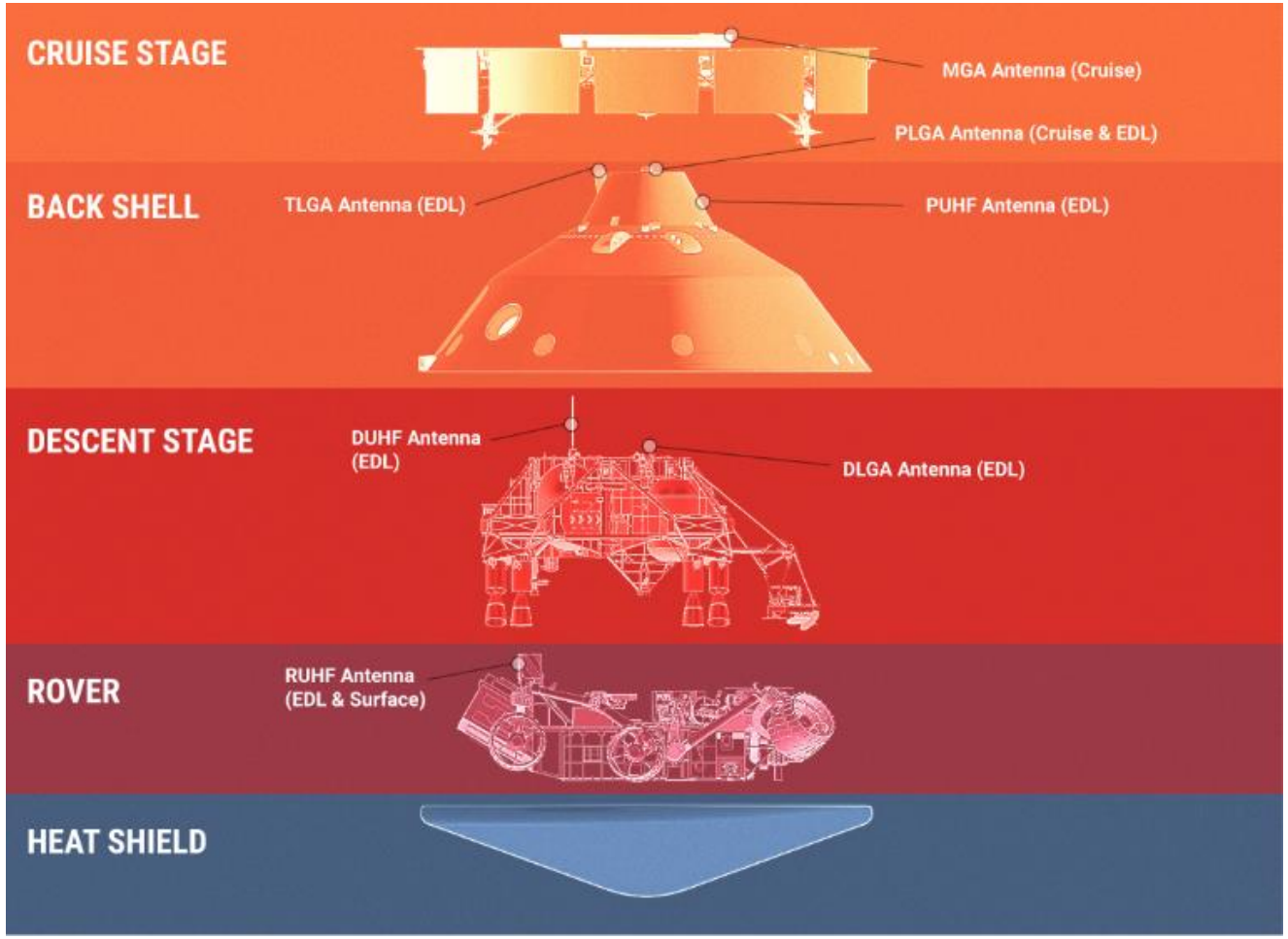
High gain: more link performance,
more pointing dependence.

Low gain: broader coverage,
lower achievable rate.

Two low-gain antennas can reduce
spacecraft blockage.

Image: NASA/JPL-Caltech [1]

Perseverance: Antennas Change With Mission Phase



Cruise

Entry, descent, and landing

Surface operations

The useful RF path changes as the vehicle configuration and geometry change.

Image and mission facts: NASA/JPL-Caltech [7]

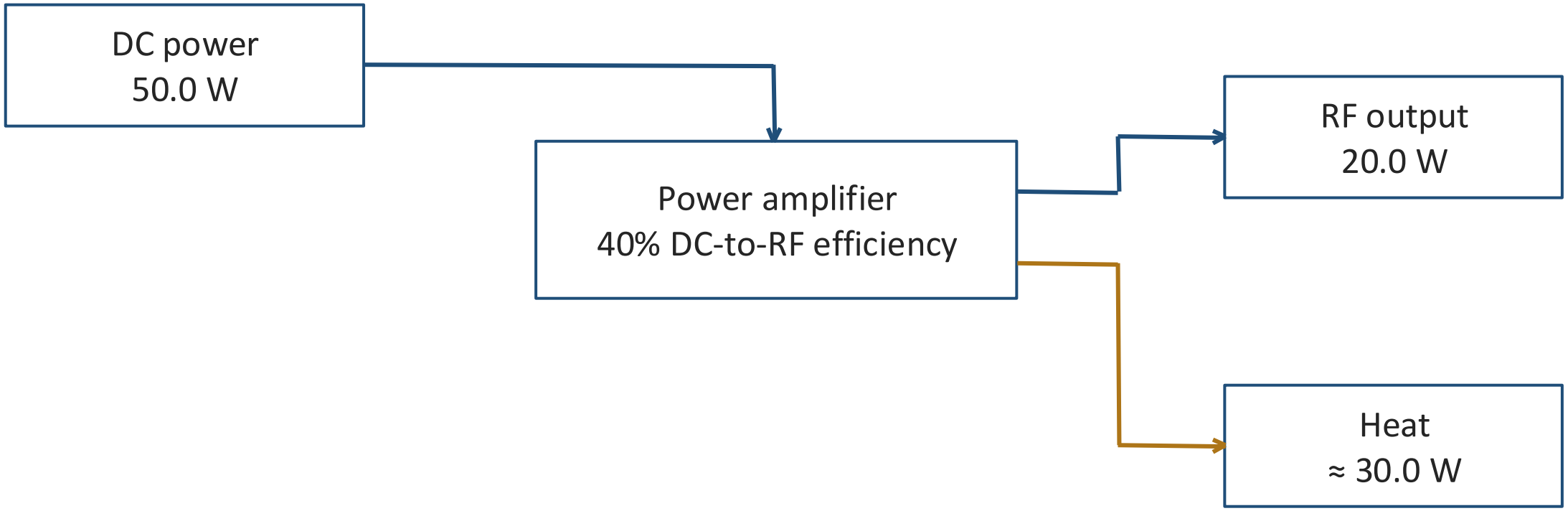
Perseverance: Different Antennas, Different Services



UHF antenna	X-band high gain	X-band low gain
Relay via Mars orbiters	Pointed direct-to-Earth link	Broad-coverage direct link

Image: NASA/JPL-Caltech [7]

Power Amplifiers Also Consume Spacecraft Power

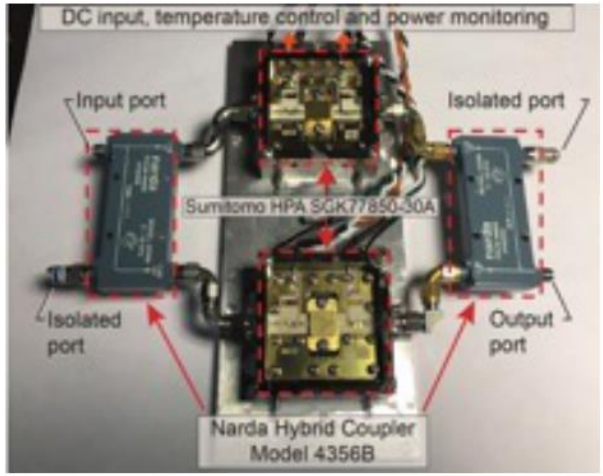


$P_{DC} = P_{RF} / \eta = 20.0 / 0.40 = 50.0$ W

$P_{heat} \approx P_{DC} - P_{RF} = 50.0 - 20.0 = 30.0$ W

Teaching example • Efficiency excludes other spacecraft loads

SSPA and TWTA: Both Remain Relevant



Solid-state power amplifier

Semiconductor devices, including GaN.
Compare linearity, efficiency,
power, radiation tolerance and heritage.

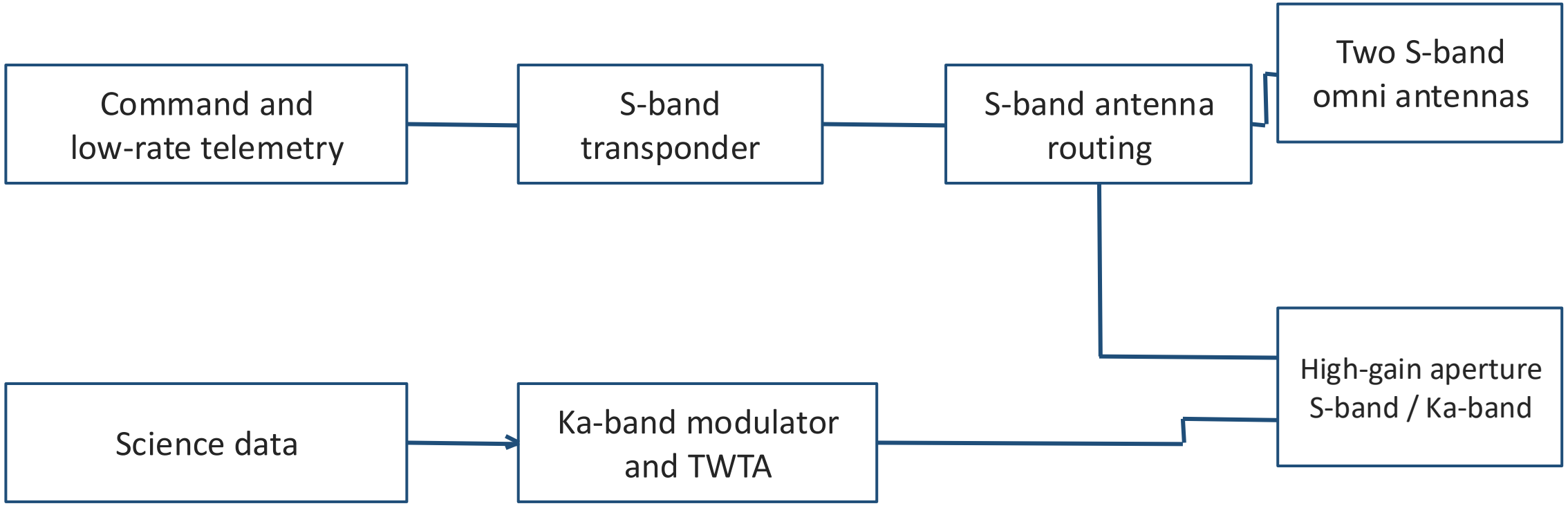


Traveling-wave tube amplifier

Electron beam transfers DC energy to RF.
Useful at high microwave power.
Includes a high-voltage power conditioner.

Images and technology comparison: NASA Glenn, R. N. Simons [8]

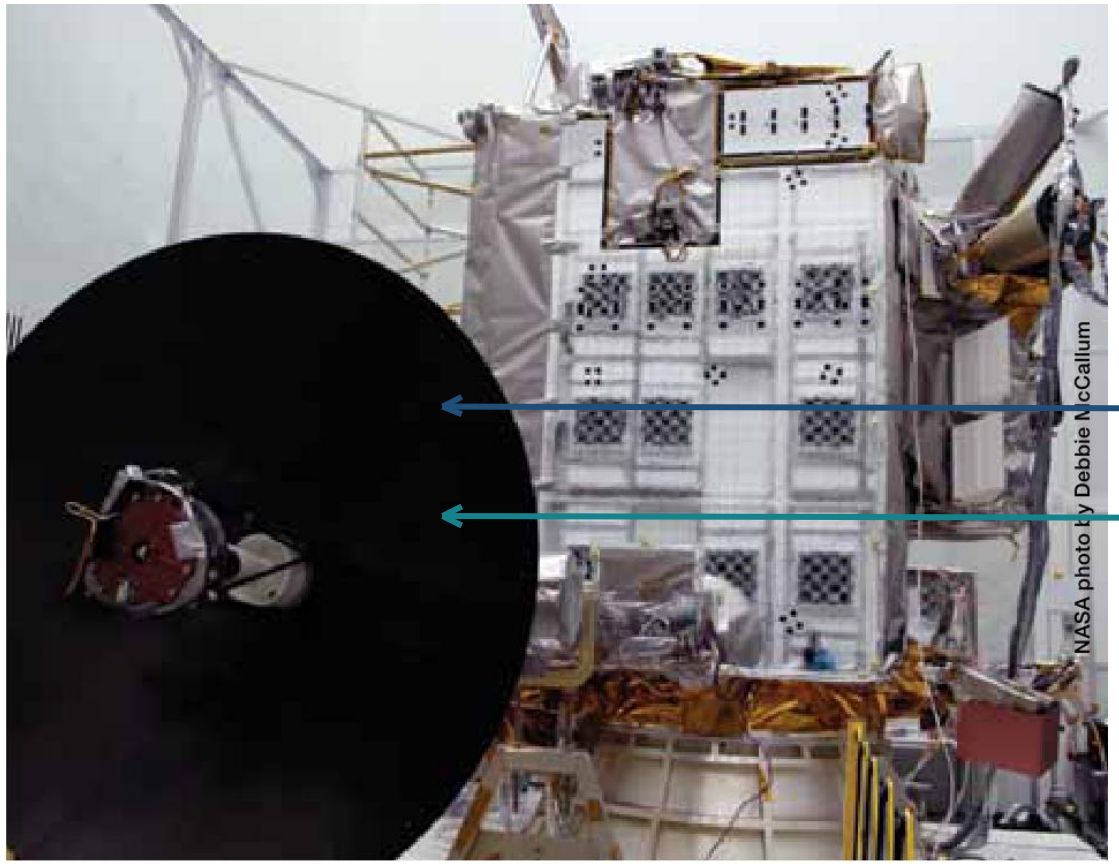
LRO: Separate Science and Command Functions



One spacecraft can need several RF paths and more than one antenna.

NASA PDS LRO Radio Science Subsystem [9]

LRO: Two Bands Share the Flight Antenna



Shared S-/Ka-band high-gain antenna

S-band
command / telemetry

Ka-band
science downlink

Separate RF feeds use one reflector and one pointing assembly.

Ka-band has the narrower beam.
A pointing failure can affect both bands.

Photographs: NASA/GSFC [35]. Functional paths: [9, 23].

Break

Next: what still works when something fails?

Redundancy Applies to a Complete Function



A command path also depends on power, cables, switching, control logic, and the ability to select the surviving path.

Two radios do not establish redundancy if one shared failure disables both.

[10]

Example C: Question: Two Radios, One Antenna

Choose one TX. Keep both RX inputs connected.

Transponder A
TX and RX

Transponder B
TX and RX

RF connections
to be designed

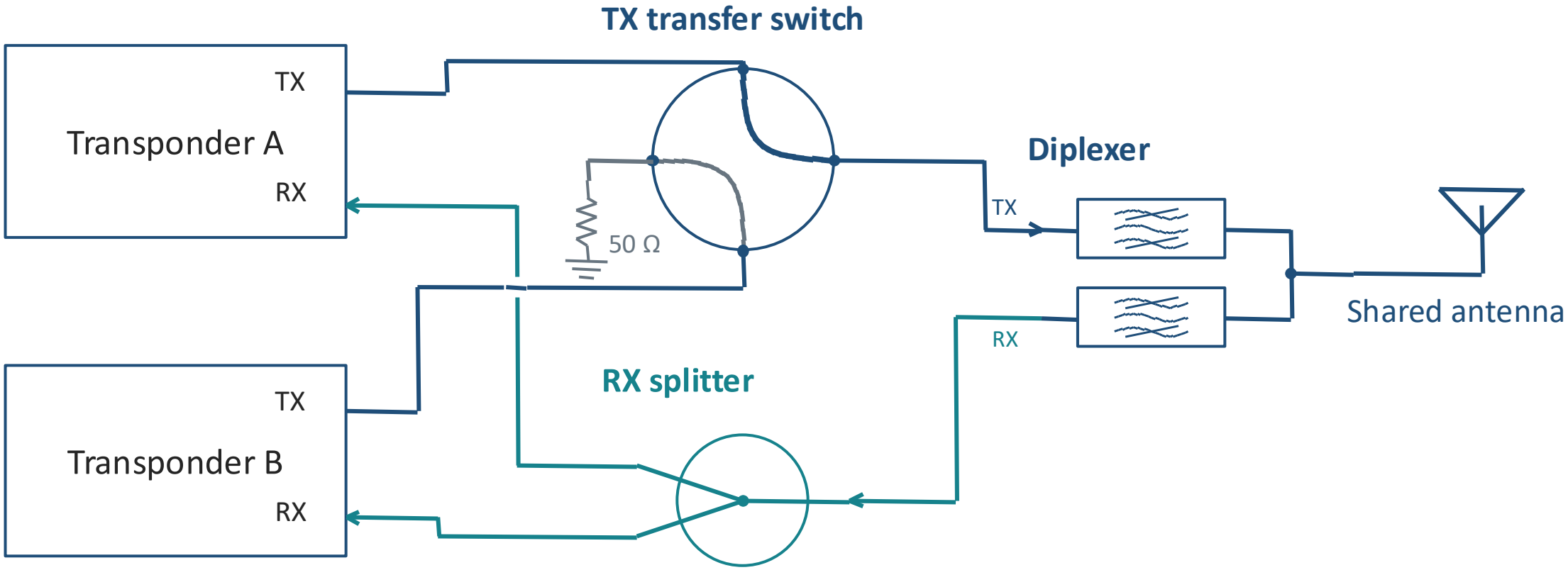


Shared antenna

Where do the switch, splitter, and diplexer belong?

Teaching example

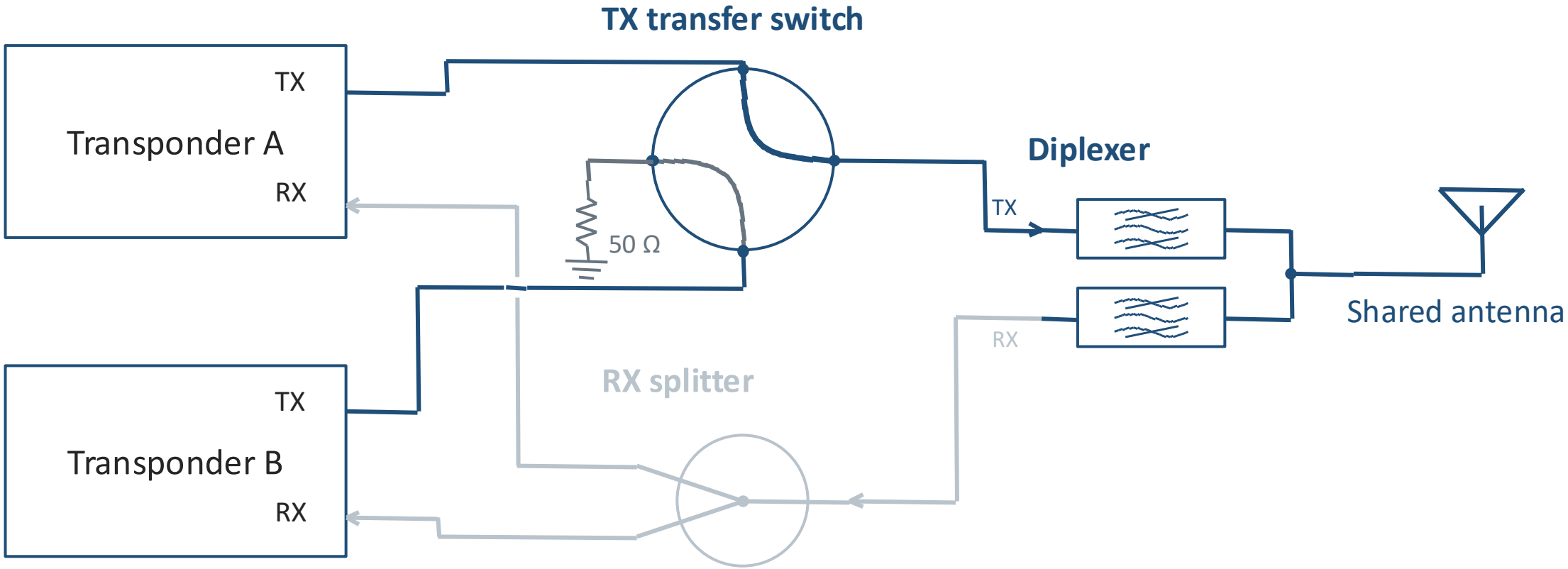
Example C: Worked Answer: The RF Connections



One transmitter connects to the antenna. Both receivers listen.

Teaching topology • Each selected path includes cable, switch, diplexer and splitter loss.

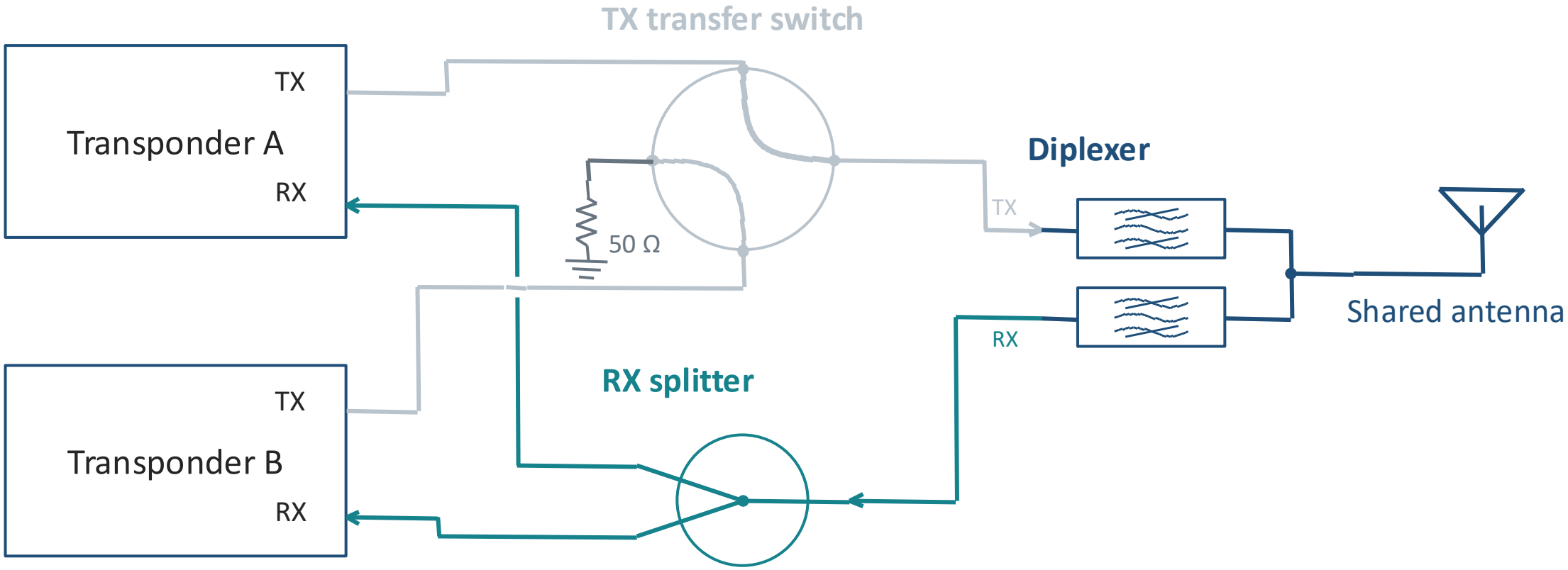
Example C: Normal Transmit Path



The switch directs the active transmitter's power into one RF path.

Teaching topology • Each selected path includes cable, switch, diplexer and splitter loss.

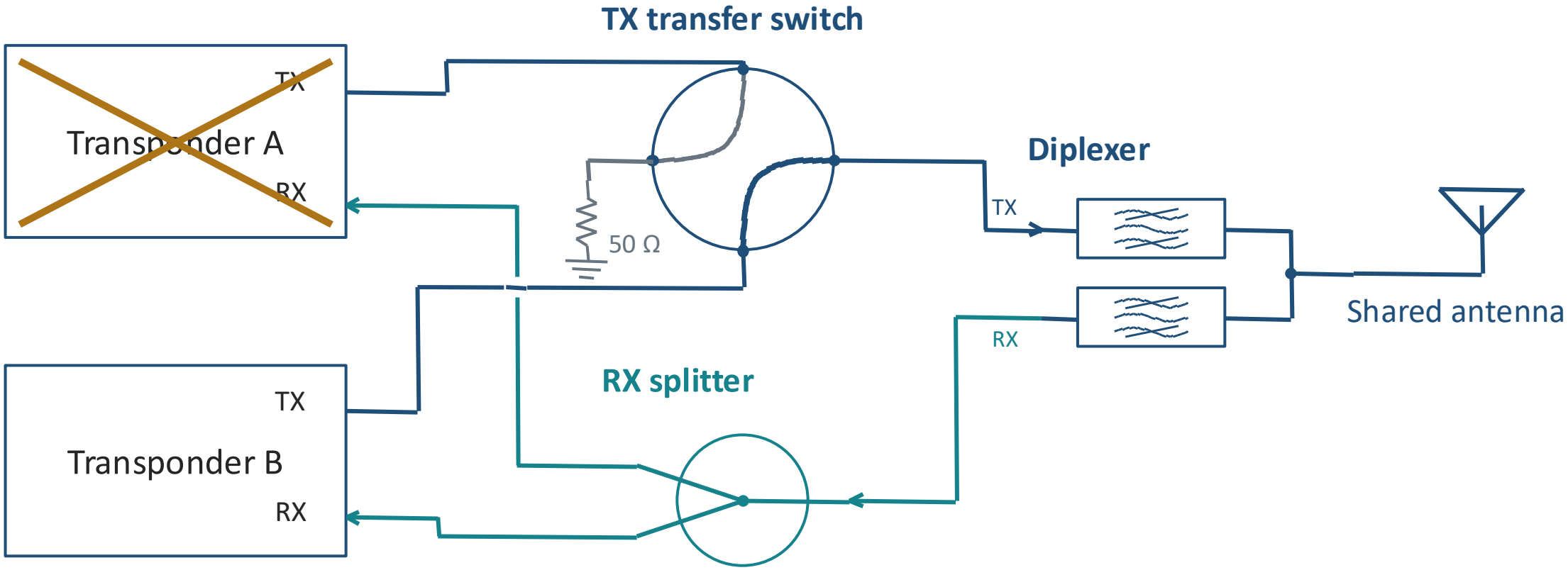
Example C: Command Reception



Command reception does not require switching to the other receiver.

Teaching topology • Each selected path includes cable, switch, diplexer and splitter loss.

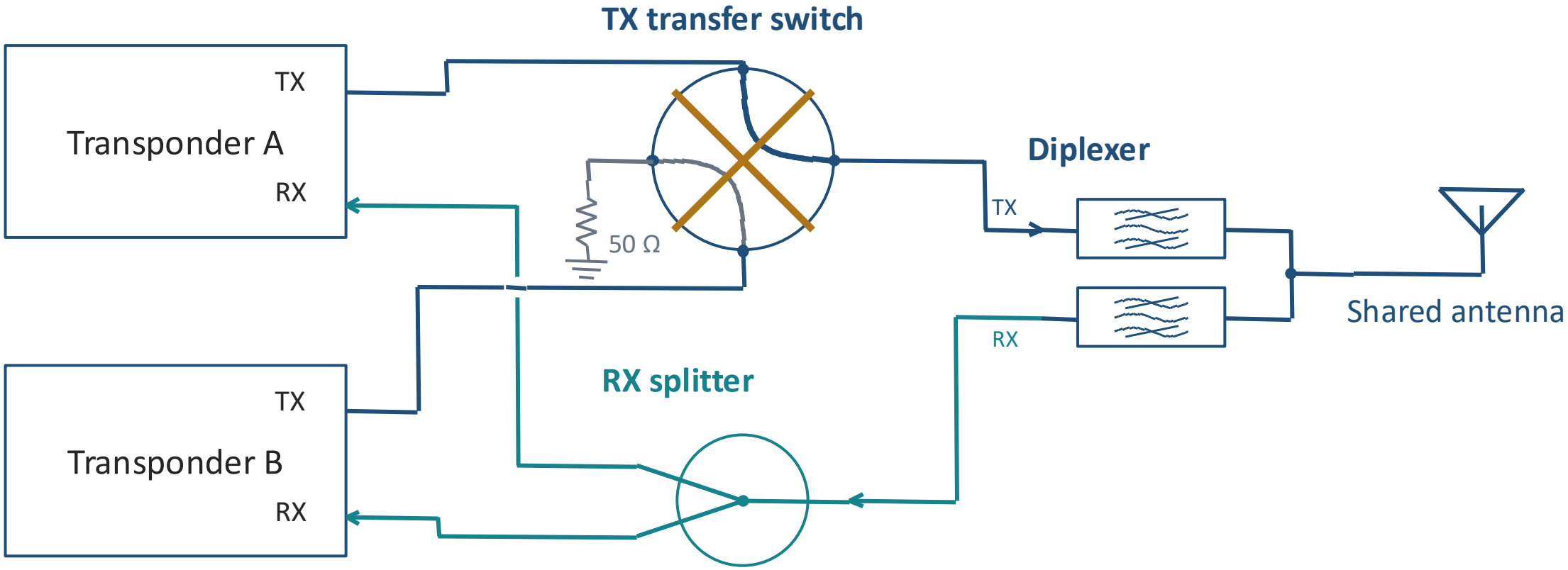
Example C: Transponder A Fails



After failover: TX B reaches the antenna. RX B still accepts commands.

Teaching topology • Each selected path includes cable, switch, diplexer and splitter loss.

Example C: Question: The TX Switch Fails Open



Can the second transmitter restore the downlink through this switch?

Teaching example • Fault: the switch antenna-side port becomes an open circuit.

Example C: Worked Answer: Shared Hardware

No. Both transmitters depend on the same antenna-side switch port.

The receiver paths may still accept commands.

A spare transmitter does not bypass an open RF connection.

A separate path can improve fault coverage, but adds hardware.

Name the failure that the design tolerates, and the failures it does not.

[10]

Recovery Must Be Possible Without the Lost Link



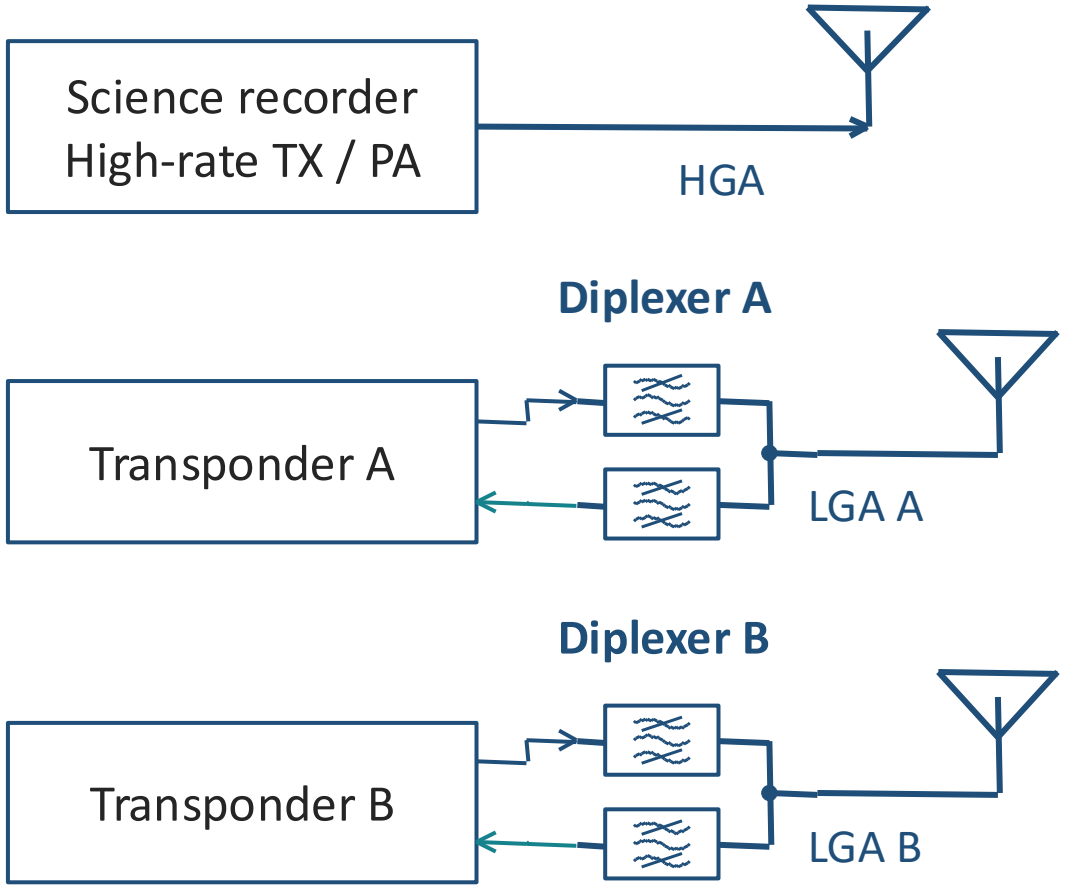
A timer or fault detector can request a radio/path change.

The recovery logic needs surviving power and control interfaces.

A recovery command cannot depend entirely on the failed receiver.

NASA fault-protection guidance [10]

Example D: Operating Modes and Their RF Paths



SCIENCE RETURN

High-rate TX → HGA.
Receiver A or B keeps listening.

SAFE MODE

Low-rate A → LGA A.
Keep command receivers listening.

A CHAIN LOST

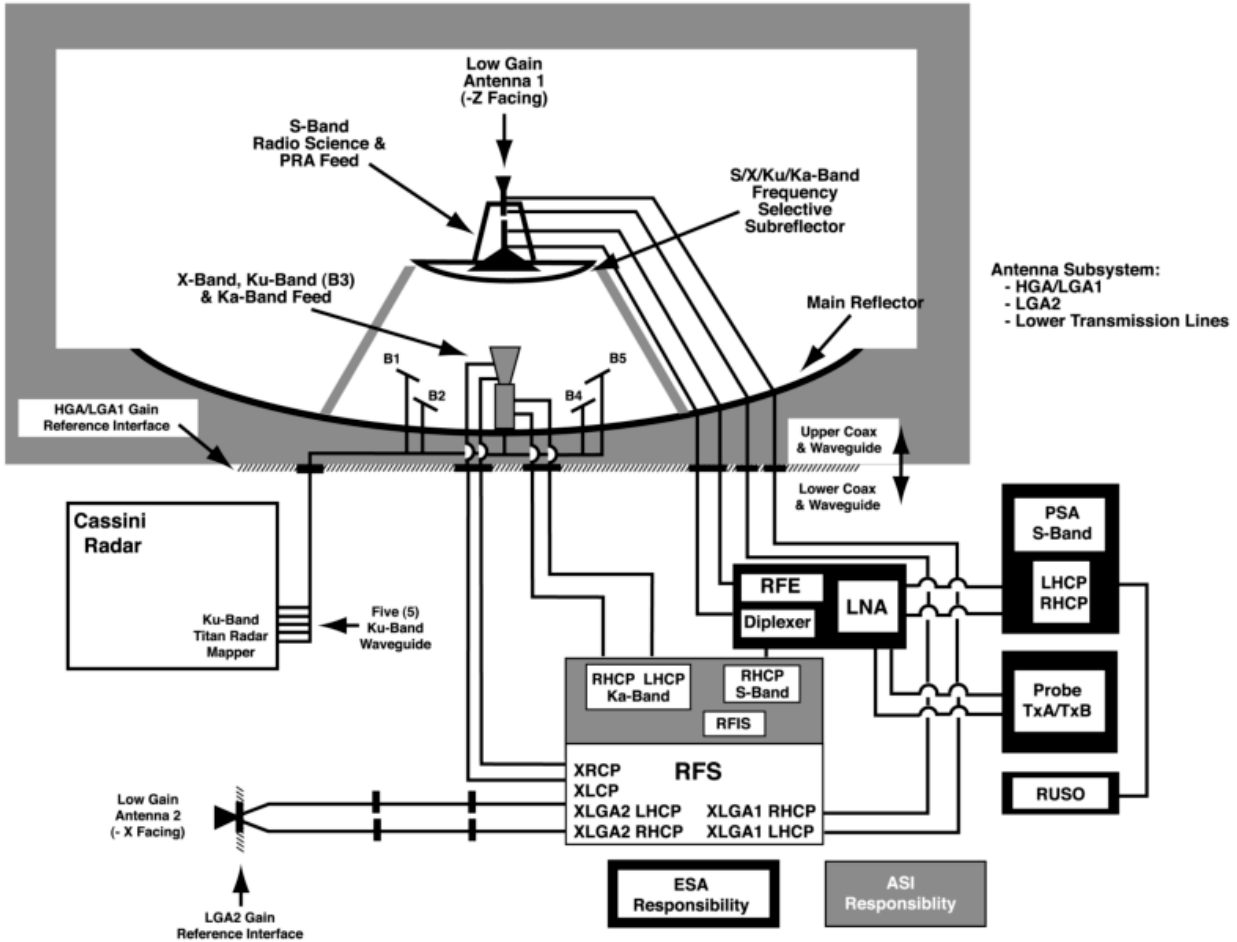
Use B and LGA B for TX and RX.

HGA POINTING LOST

Use a low-rate path within coverage.

Teaching architecture • Include each route's RF loss. Check coverage, power and link margin.

Cassini: One Spacecraft, Several RF Services



X-band: command and telemetry

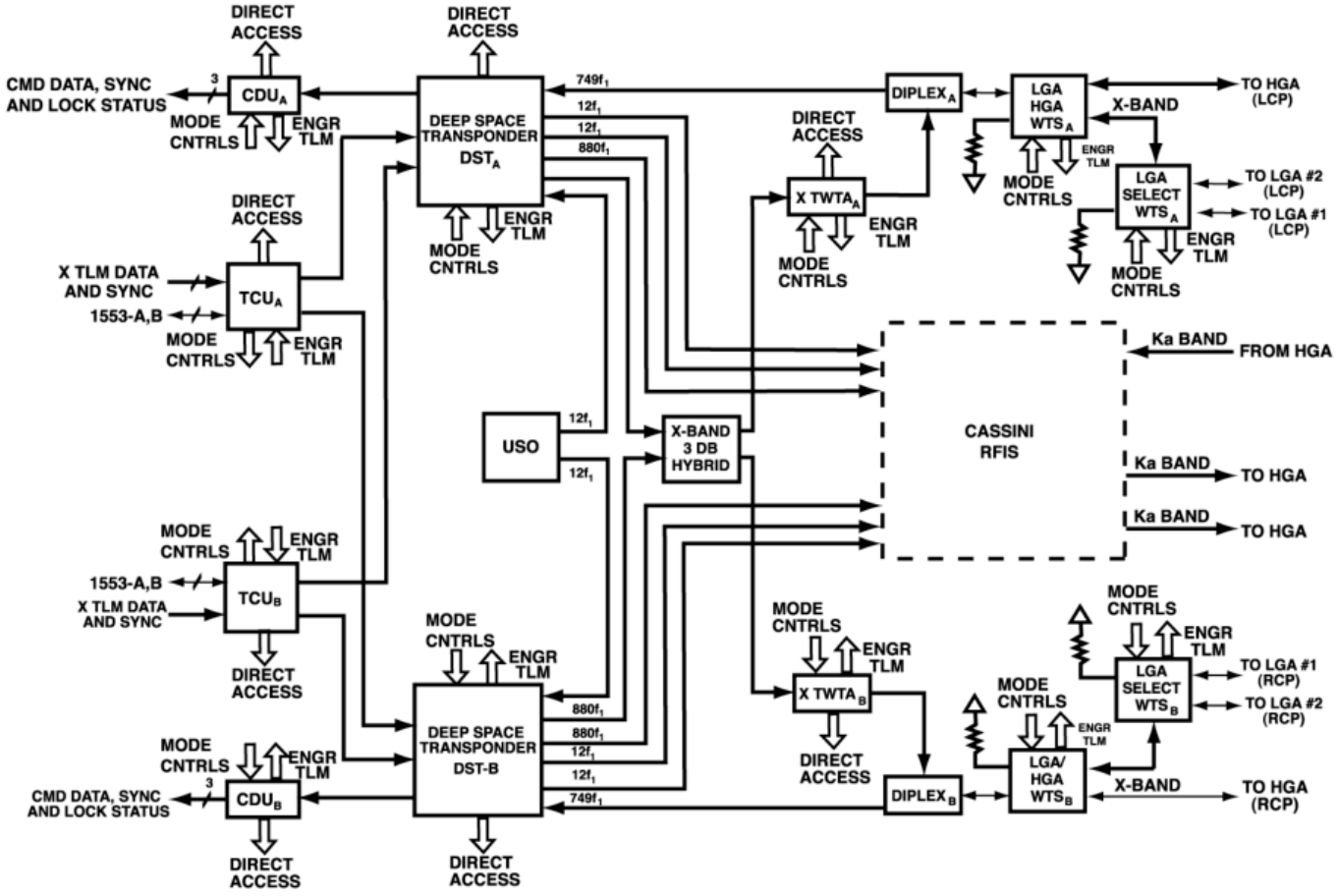
S / Ka: radio science
S-band: Huygens relay

Ku-band: radar

HGA + two X-band LGAs

Image: NASA/JPL, DESCANSO 3, Fig. 3-2 [21]; mission context [22].

Cassini: Normal and Alternate RF Paths



NORMAL PAIRING

DST A → TWTA B

IF TWTA B FAILS

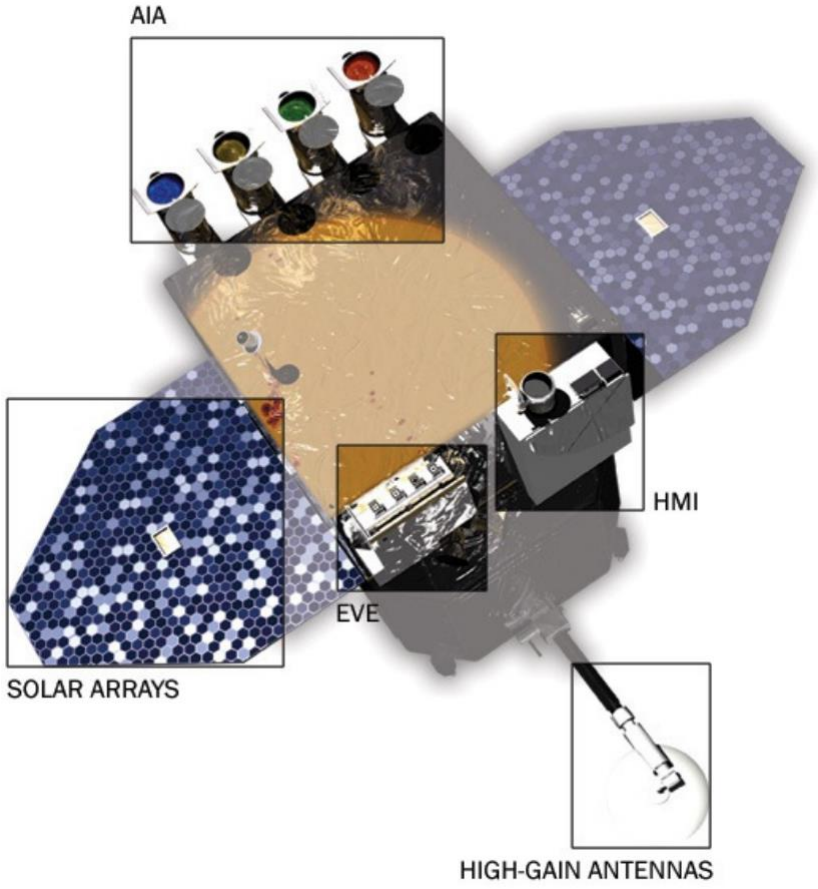
DST A → TWTA A

The hybrid connects either exciter to either amplifier.

Match the downlink polarization on Earth.

NASA/JPL, DESCANSO 3, Fig. 3-4 [21]. WTS = waveguide transfer switch.

SDO: Communications Shapes the Spacecraft



The observatory points at the Sun.

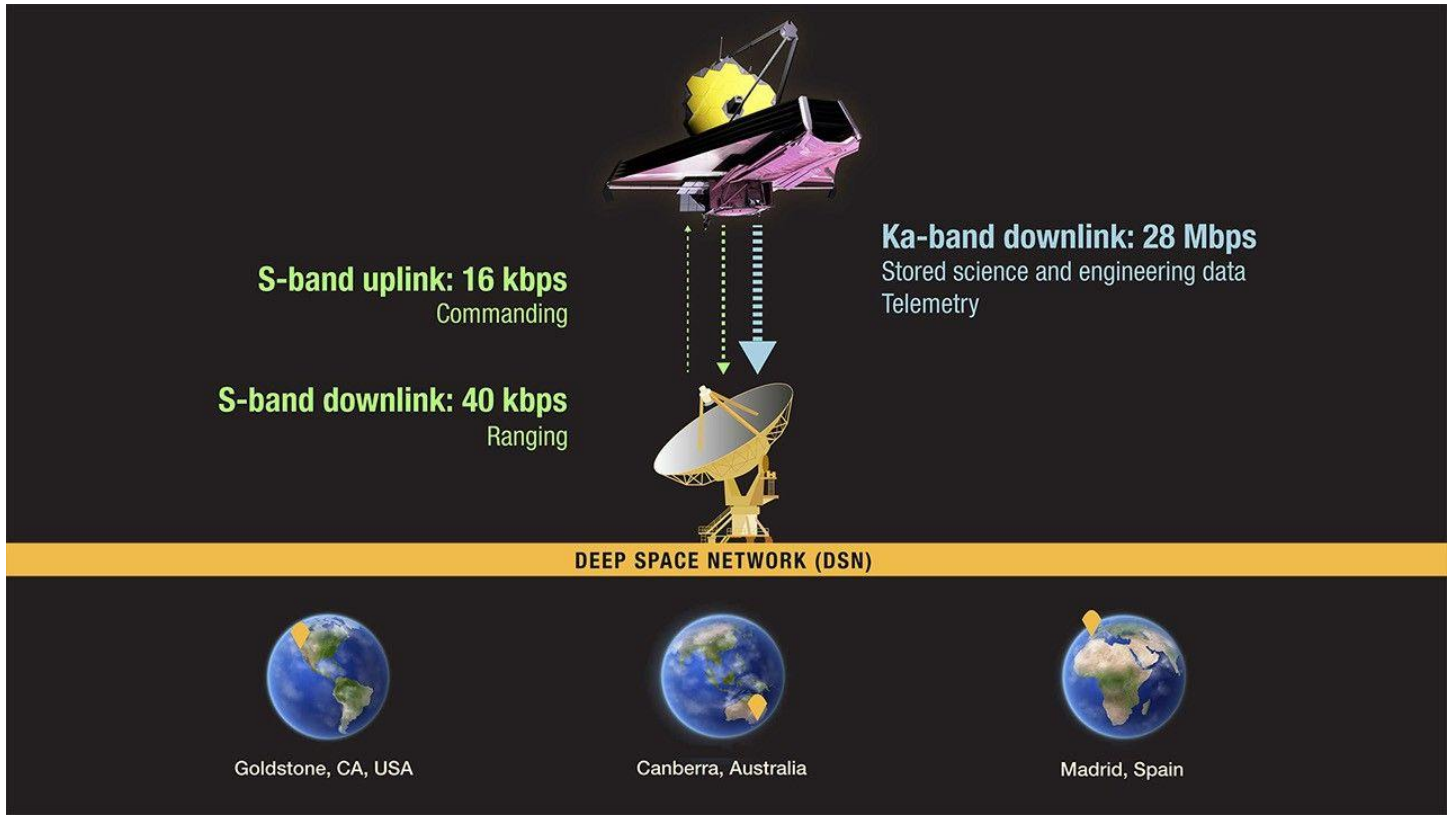
Steerable antennas track Earth.

Solar-array shape avoids blocking the antenna view.

Science downlink: 130 Mbit/s at about 26 GHz.

Image and facts: NASA/GSFC SDO press kit, p. 6 [13]

Webb: Science Data and Command Use Different Bands



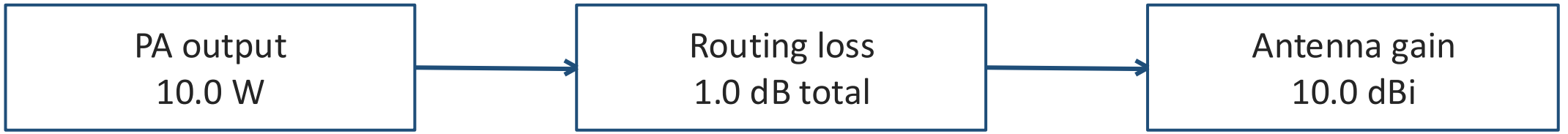
Ka-band HGA
Science downlink

S-band MGA
Command and telemetry

A shared mission can require different RF services and antenna gains.

Illustration: NASA, ESA, CSA, Pam Jeffries (STScI) [14]

Example E: The Losses Must Follow the Actual Route



$L_{route} = 2(0.1) + 0.5 + 0.3 = 1.0$ dB

$[EIRP] = [P_{TX}] - L_{route} + [G_{TX}]$

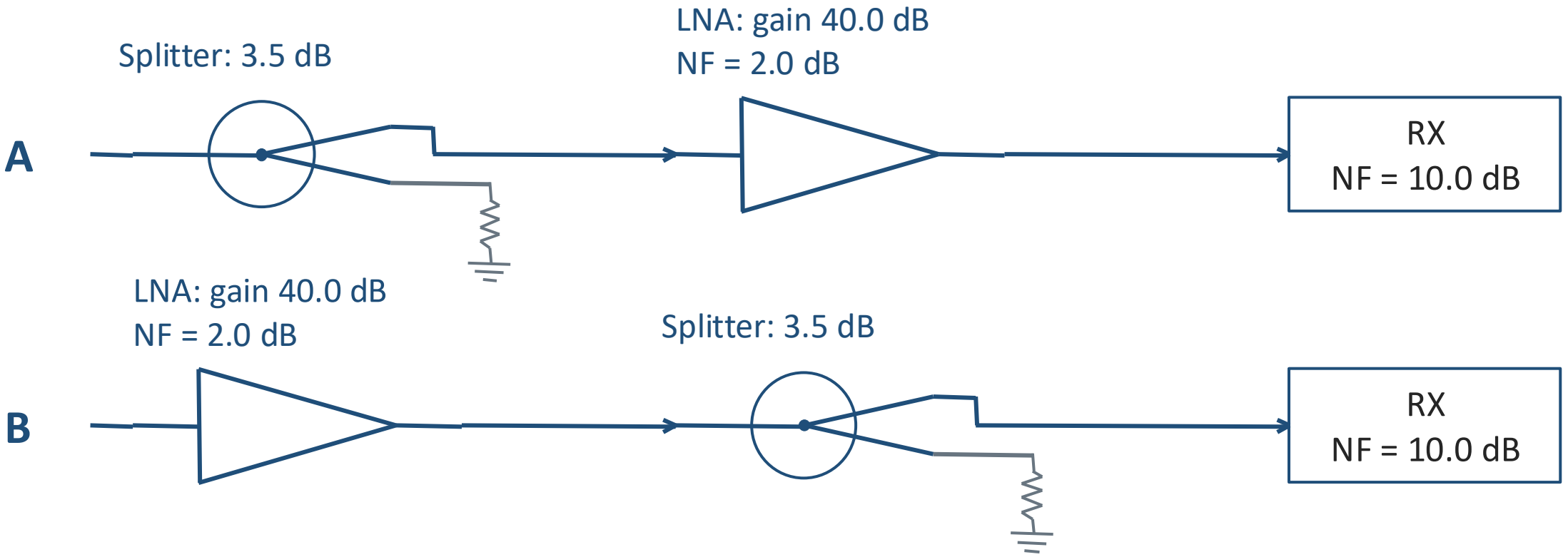
$[EIRP] = 10.0 - 1.0 + 10.0 = 19.0$ dBW

Two cables + one switch + one diplexer. Do not subtract these losses twice.

Teaching values • Antenna gain referenced to its feed terminal.

Example F: Question: Where Should the Splitter Go?

Which arrangement gives the lower overall noise figure?



Teaching example • Splitter: 3.0 dB division + 0.5 dB excess. Passive parts at 290 K.

Example F: Worked Answer: Noise Figure and Redundancy

$F_{total} = F_1 + (F_2 - 1)/G_1 + (F_3 - 1)/(G_1G_2)$

Splitter: F = 2.239, G = 0.447. LNA: F = 1.585, G = 10,000.

A: $F_{total} = 2.239 + (1.585 - 1)/0.447 + 9/(0.447 \times 10{,}000)$

B: $F_{total} = 1.585 + (2.239 - 1)/10{,}000 + 9/(10{,}000 \times 0.447)$

NF = 10 $\log_{10}$(F): A = 5.5 dB B = 2.0 dB

B has lower NF, but the shared first LNA is a common failure point.

[6] • Splitter total loss 3.5 dB. Other port matched at 290 K.

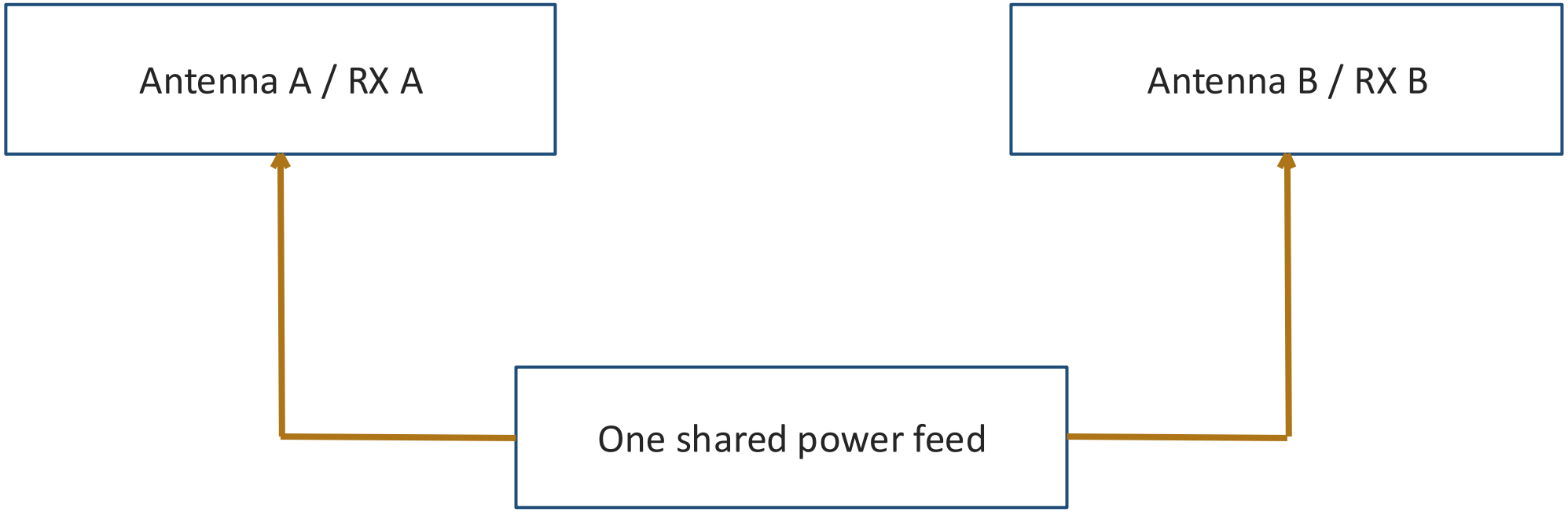
Radio Selection Includes More Than Data Rate

Check	Why it matters
RF band, power, NF and interfaces	The radio must fit the physical link and spacecraft
Modulation, coding and data rates	Both ends must support the same modes
Radiation, temperature and flight heritage	A laboratory specification is not flight qualification
Software-defined functions	Reconfiguration helps only within hardware limits
Power states and recovery behavior	A spare must be usable after the assumed failure

NASA Small Spacecraft State-of-the-Art, Communications [15]

Question: What Else Must Survive?

The RF paths are separate. What common failure can still defeat both?



Power connections shown in gold.

Teaching example

Worked Answer: Independence Extends Beyond RF

A failure of the shared power feed can disable both receivers.

Independent, protected feeds can improve the intended fault coverage.

Also check shared decoders, control lines, connectors and software.

A credible design identifies both the surviving path and the limits of its redundancy.

[10]

What a Complete Communications Design Must Show

Evidence	Question it answers
Operating concept and modes	When and how does the spacecraft communicate?
Hardware diagram	What equipment and connections must fly?
Path-specific link budgets	Does each required link have enough margin?
Failure-response table	What survives, and how do we select it?
Trade rationale	Why is this combination appropriate?

The Hardware Choices Return to the Link Equation

$$[S/N_0] = [EIRP] - L_p - L_{atm} - L_{point} - L_{pol} + [G/T] + 228.6$$

$$[EIRP] = [P_{TX}] - L_{TXroute} + [G_{TX}]$$

$$[G/T] = [G_{RX}] - [T_{sys}]$$

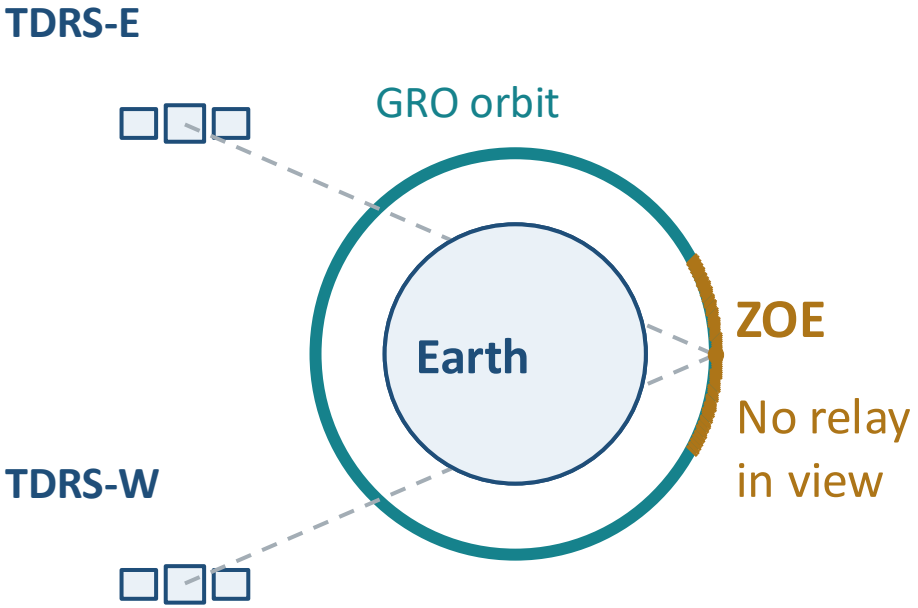
$$[E_b/N_0] = [S/N_0] - 10\log_{10}(R_b / 1\ \text{bit/s})$$

TX routing changes EIRP. RX routing changes system noise.
Antenna choice and pointing change the usable link in each mode.

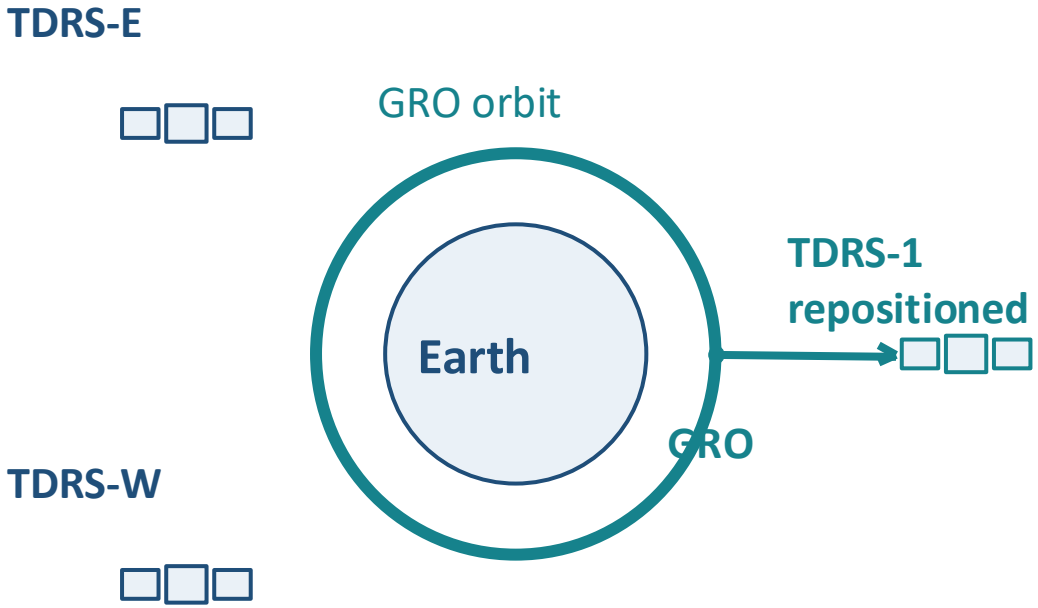
[6, 12] • Loss terms are positive dB quantities.

GRO: Closing the Gap in Relay Visibility

BEFORE: a relay-visibility gap



AFTER: TDRS-1 covers that gap



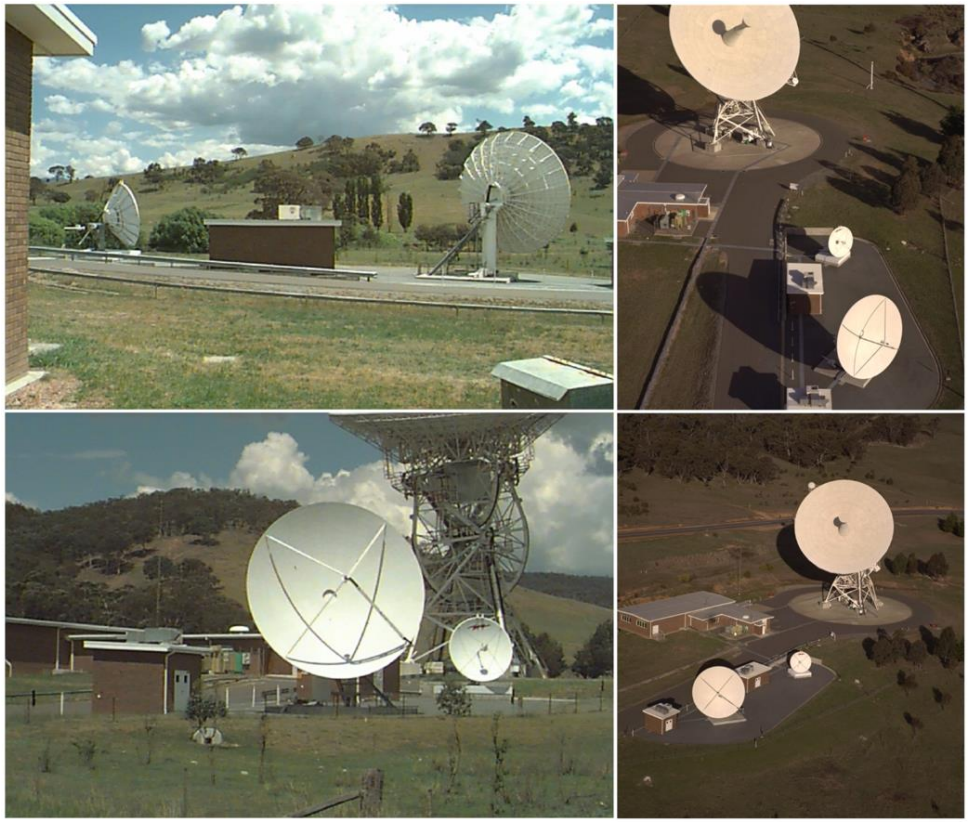
Green: a relay is in view.
Amber: no relay is in view.

At least one TDRS in view through the full orbit

TDRS-1 could no longer see White Sands. Canberra supplied its ground connection.

Conceptual redraw from the GRTS engineering history [32]. Not to scale or exact orbital geometry.

GRTS: Closing the Relay Network's Coverage Gap



1. Move TDRS-1 to close the ZOE.

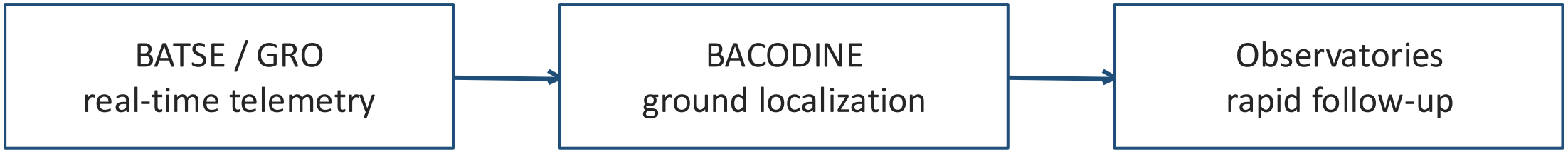
2. TDRS-1 is now outside White Sands' field of view.

3. Add Canberra ground capability for the repositioned relay.

Relay-network reconfiguration enabled full-orbit coverage.

Images/history: NASA [31]; Zillig and Valvano [32]. ZOE = Zone of Exclusion.

BATSE → BACODINE → GCN → Swift



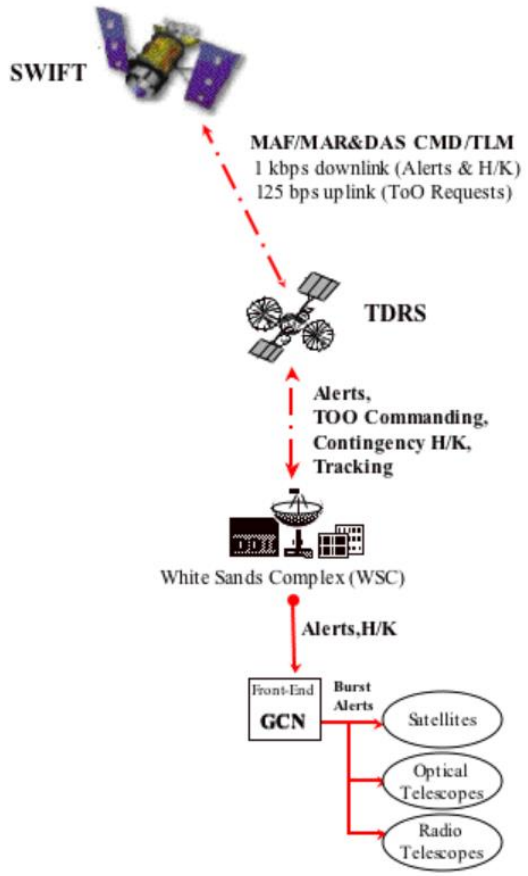
1993: use the live telemetry stream to distribute burst positions.



Swift also repoints its own X-ray and UV/optical telescopes autonomously.

NASA GCN history [33]; original BACODINE description [34]; Swift [24].

Swift: Communications as Part of the Discovery



A gamma-ray burst is unpredictable.

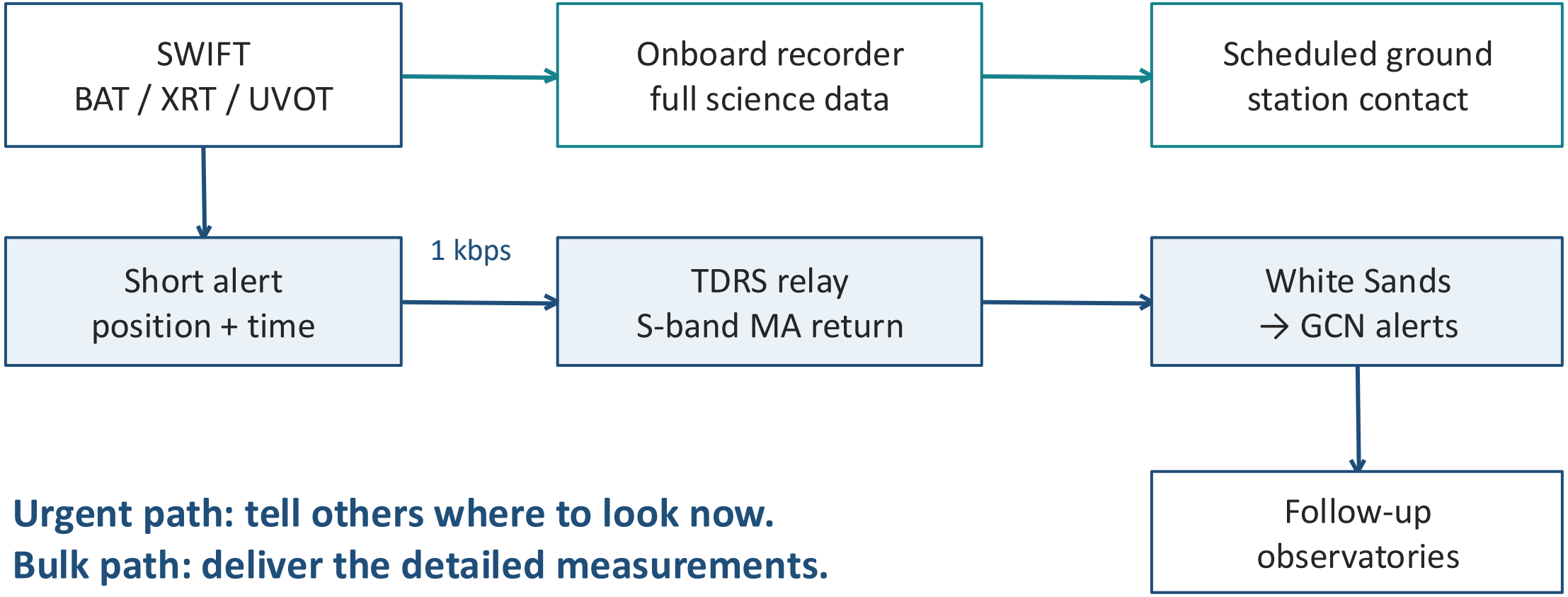
Its afterglow begins fading immediately.

Swift points its own telescopes
and alerts the rest of the world.

The mission has ended. The architecture remains a powerful lesson.

NASA/GSFC mission-flow illustration [24]; mission status [28].

Swift: Two Data Paths, Two Different Jobs



Urgent path: tell others where to look now.
Bulk path: deliver the detailed measurements.

Functional redraw from NASA Swift [24, 26] and SpaceOps [25].

Swift: Multiple Access and Demand Access

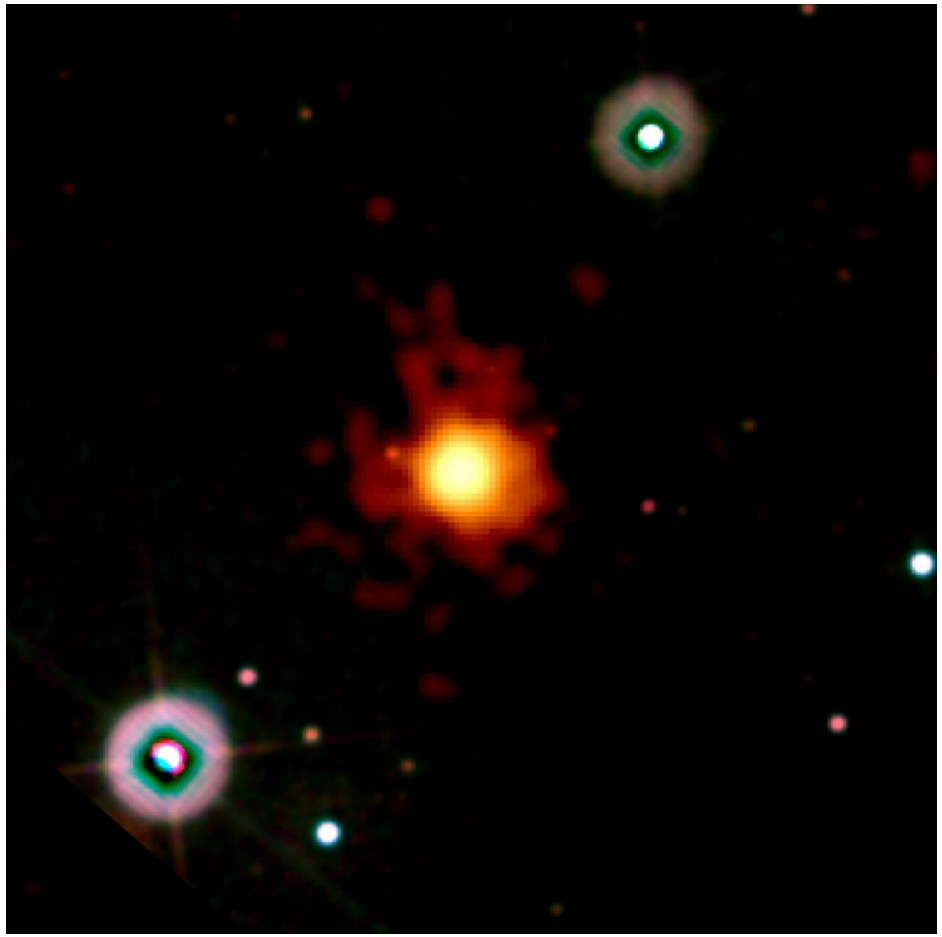
MULTIPLE ACCESS	DEMAND ACCESS
Several spacecraft can share the TDRS relay service.	A burst alert can use the relay without waiting for a routine preplanned ground pass.

Question: Which matters most for the first burst alert?

Answer: low delivery delay. A short message can start observations while the transient is still bright.

[24–26] • Access requires a visible relay with available capacity.

Swift Discovery: A Burst from the Young Universe



23 April 2009: Swift finds GRB 090423.

Rapid position alerts guide follow-up.

Ground infrared observations: $z \approx 8.2$.

A massive star died when the universe was only about 600 million years old.

Fast communication turned a fleeting event into a measurable discovery.

Image: NASA/Swift/Stefan Immler [27]. Science: ESO [29]; event TDRSS archive [30].

Next: A Closed Link Is Not Enough

The RF hardware must provide a usable path.

The link budget must support the information rate.

The network must provide access when the information matters.

How can many spacecraft share a relay
and still deliver urgent information promptly?

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[25] Sobchak, Shinners, Shaw. NASA Space Network Project Operations Management. SpaceOps 2018, Fig. 7.

[26] NASA Swift Burst Advocate Guide, v0.8. TDRSS and ground links.

[27] NASA Imagine the Universe. Swift finds GRB 090423 (2009).

[28] NASA Swift mission-end update (23 October 2026); Aerospace reentry prediction (checked 1 November).

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[30] NASA Swift BAT archive. GRB 090423 TDRSS data products.

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[31] NASA. March 1994: GRTS Declared Operational.

[32] D. J. Zillig and J. Valvano. The GRO Remote Terminal System. SpaceOps 1994; NASA NTRS 19950010773.

[33] NASA. History of GCN.

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